The House Always Wins: Self-Preferencing After Google Shopping

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Abstract

Self-preferencing by large digital platforms has become a central concern for antitrust authorities and policymakers. We study one of the earliest and most prominent cases of self-preferencing, the European Commission's 2017 Google Shopping decision. Google was accused of systematically favoring its own comparison shopping service in search results while downgrading rival comparison shopping websites in its search ranking. The Commission prohibited this conduct but left Google discretion over how to comply: following the €2.42 billion fine, Google redesigned its search results page (SERP), reconfiguring the Shopping Unit to let third-party comparison shopping services (CSS) participate in its advertising system. Because the decision constrained Google's conduct without prescribing the design of the remedy, what can be causally identified is the effect of Google's redesign itself, rather than the effect of the decision in the abstract. Using web traffic data, we estimate the effect of this redesign on traffic to comparison shopping websites, comparing European countries where it was implemented with non-European markets where no equivalent design change occurred. Event-study evidence shows that organic traffic to comparison shopping websites in Europe declined further relative to non-European control markets after the redesign, while advertising-related traffic does not significantly increase. Notably, smaller CSS platforms experienced disproportionately larger losses in organic traffic, while larger competitors managed to sustain their visibility. These results suggest that Google's redesign, while formally compliant, did not restore the visibility of rival CSS – an unintended consequence consistent with a remedy that regulators did not sufficiently monitor or constrain at the level of interface design. While previous work has discussed potential limitations of Google's compliance mechanism, our paper provides the first causal empirical evaluation of its effects.

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1 Introduction

Since the adoption of the Digital Markets Act (DMA), concerns about self-preferencing by digital gatekeepers have become a defining feature of European competition policy. Yet the intellectual and legal foundations of the DMA can be traced back to the landmark 2017 decision of the European Commission in the Google Shopping case. In that case, the European Commission imposed a €2.42 billion fine on Google and required the company to cease the conduct immediately. Yet the decision itself only prohibited the conduct; it did not prescribe how Google should comply, leaving the design of the remedy to Google's own discretion. Despite the importance of the decision and its far-reaching implications, empirical evidence on whether the resulting redesign effectively restored competition remains scarce.

We examine whether the design changes Google introduced in response to the prohibition of self-preferencing restored traffic to rival CSS. To answer this question, we employ a dynamic difference-in-differences design to assess the effects of the compliance mechanism Google introduced in September 2017, following the European Commission's 2017 decision against Google. The Commission concluded that Google leveraged its dominance in general search to reduce the visibility of rival CSS in organic rankings, while simultaneously granting preferential placement to its own service through a prominently displayed results box. In response, Google introduced a new mechanism in September 2017 that allowed third-party CSS to participate within its Shopping unit. This mechanism opened access to a previously privileged space, but did not alter Google's control over ranking and interface design in general search. Concretely, rival CSS could now have their product listings included in the Shopping Unit, but only as a small label beneath the product image; the primary clickable element continued to direct users to the merchant rather than to the CSS itself (Section 3 details this redesign, including click-through evidence). Ex ante, this design feature limits how much of the Shopping Unit's traffic can reach rival CSS even when they participate, regardless of whether the remedy is otherwise effective. Two further features of the redesign are relevant for interpreting our results. First, because the Shopping Unit is displayed above organic results, its prominence – independent of who appears within it – may reduce how far users scroll before reaching organic CSS listings, potentially depressing organic traffic even for CSS that do not participate in the Shopping Unit at all. Second, because participation in the Shopping Unit requires bidding through Google's advertising system, any gain in visibility comes at a cost and is conditional on paid participation, rather than being restored through organic ranking. These three features – the redirect to merchants, the crowding-out of organic results, and the shift toward paid participation – map directly onto the three outcomes we examine below: paid traffic, organic traffic, and ranking position. Taken together, they give reason to expect, ex ante, that the redesign would not straightforwardly restore traffic to rival CSS; whether this expectation holds in the data, and how large the resulting effects are, is the empirical question we turn to next. Because Google retained full control over the design of this mechanism, our estimates should be interpreted as the causal effect of the specific design Google chose to adopt, rather than of the Commission's decision per se.

The competitive significance of this mechanism depends on the market power and reach of the platform that controls it. At the time of the decision, Google already held over 90% of global search market share and served approximately 3.0 billion users worldwide (Alphabet

Inc., 2018; StatCounter Global Stats, 2017). Since then, while market share has remained broadly stable, the scale of activity and monetization has expanded substantially. Average daily searches increased from around 4.7 billion in 2017 to approximately 10.8 billion in 2024 (Land, 2024), while Search revenues rose from $69.8 billion to $198 billion over the same period (Statista). The number of users also grew from roughly 3.0 billion to 4.35 billion (SimilarWeb), reinforcing the scale and persistence of Google's dominance. Given this central role as the primary gateway to online information and commerce, visibility within Google's search results is economically critical for downstream firms. CSS (CSS), in particular, depend heavily on referral traffic from general search to attract users and generate revenue. In such an environment, even small changes in ranking or design can translate into substantial shifts in traffic allocation and competitive outcomes.

The European Commission's investigation documented a sustained decline in organic traffic to CSS in Europe following Google's 2011 Panda update, continuing through 2016. We examine whether the fine and the subsequent implementation of Google's redesigned Shopping Unit altered this trend. We find that they did not: after the fine and the rollout of the compliance mechanism, organic traffic to rival CSS in Europe continued to decline relative to non-European control markets, where no equivalent redesign took place. Moreover, although the remedy formally allowed third-party CSS to participate in the Shopping Unit, we find no significant overall increase in advertising-related traffic. A heterogeneity analysis reveals a marked asymmetry across platforms. Larger CSS experience smaller losses in organic traffic than smaller CSS. However, even for larger platforms we do not observe any meaningful increase in paid traffic after the remedy. To explain the absence of changes in paid traffic, we explore mechanisms that may limit the effectiveness of the remedy, including the design of the Shopping Unit.

The 2017 decision against Google highlights a broader challenge in enforcement and remains a foundational precedent for the DMA. Compliance assessment demands active evaluation of the frequent design changes implemented by gatekeepers, as well as clear standards for distinguishing legitimate product design from discriminatory conduct (Crémer and de Streel, 2023). This evaluation in platform markets is inherently complex, particularly when changes involve algorithmic design, ranking, and interface adjustments that are not directly observable. Because platform design evolves continuously, regulators must distinguish between genuine product improvements and strategic adjustments that systematically favor the platform's own services. This assessment is especially challenging when platforms operate in a dual role (Carugati, 2022; Peitz, 2023). Because the 2017 intervention did not involve systematic ex post monitoring of compliance, Google retained wide discretion over the design of its response – discretion that, as we show, it used in ways that did not restore competition in the comparison shopping market.

Our contribution is therefore best understood not as an evaluation of the Commission's decision per se, but as a causal estimate of the effect of the specific SERP redesign Google adopted in response to it. This distinction is not merely semantic: it is precisely because the decision prohibited a practice without constraining how compliance would be implemented that Google's discretion over design became consequential. Our results speak to the risks that arise when regulators prohibit conduct without also constraining, or actively monitoring, how platforms translate that prohibition into concrete design choices.

2 Literature

A growing literature studies self-preferencing by digital platforms that operate in a dual role, acting both as intermediaries and as competitors to third-party sellers. Because platforms control important mechanisms such as rankings, visibility, and access to user data, they may have incentives and the ability to favor their own services. Bougette et al. (2022) discuss several mechanisms through which this behavior may appear, including ranking manipulation, data advantages, and contractual arrangements with sellers, which can generate market distortions. Calvano and Polo (2021) situate this behavior within the broader concept of intermediation bias, whereby an attention-based platform diverts users toward the compensation-maximizing rather than the surplus-maximizing option, and cite Google's promotion of its own subsidiary websites in the Google Shopping case as an example of behavior already sanctioned by antitrust authorities.

Empirical evidence of self-preferencing has been documented. Hunold et al. (2022) show that Amazon's Buy Box design can redirect demand toward Amazon's own products by suppressing the Buy Box when third-party sellers offer relatively high prices. Using a structural model of consumer choice, Lee and Musolff (2025) estimate the impact of Buy Box prominence and show that the platform can strongly steer demand, as most consumers consider only the featured offer. While Amazon's products enjoy a prominent advantage relative to similar competing products, the authors find that this does not reduce consumer surplus because the algorithm tends to feature offers that consumers prefer. Farronato et al. (2023) and Chen and Tsai (2024) analyze ranking advantages and recommendation systems. The authors show that Amazon's algorithms favor its own products in rankings and recommendations. Closely related to our own setting, Cure et al. (2022) study Kayak, a metasearch platform for hotel bookings commercially affiliated with certain online travel agencies. They find that, for a given hotel, offers from affiliated agents are displayed more prominently than equally-priced offers from rival agents, and that Kayak's ranking incorporates revenue and affiliation considerations beyond price alone, evidence of self-preferencing in precisely the kind of price-comparison intermediary setting that CSS occupy within Google's Shopping Unit. Self-preferencing has also been studied in the context of search engines more broadly. Chiou (2017) analyzes Google's integration of Google Flights and Zagat ratings into search results. Using a difference-in-differences approach, the authors find that the competitive effects differ by type of vertical integration: Google Flights integration reduces clicks to travel agencies, while Zagat integration generates spillovers to competing review platforms. Therefore, the integration may either divert traffic from rivals or generate spillovers depending on whether the service provides price or quality information. Complementing the empirical literature, Hagiu et al. (2022) introduce a theoretical framework in which platforms both host third-party sellers and sell their own products. While the dual mode can intensify competition and increase transactions, the model shows that practices such as product imitation or self-preferencing may weaken innovation incentives and reduce price competition. The authors conclude that some remedies such as prohibiting imitation or self-preferencing can be more effective than bans on platforms selling their own products.

A related literature examines how price-comparison platforms themselves affect market competitiveness. Baye and Morgan (2001) show theoretically that an "information gatekeeper" that reduces consumers' search costs lowers the average price charged by listed sellers,

a prediction confirmed empirically using transaction data from an online price-comparison site by Baye et al. (2004). We return to this result in Section 7, where we argue it helps explain why the presence of multiple competing CSS within the redesigned Shopping Unit may itself discourage users from scrolling down to organic results.

The European Commission's 2017 Google Shopping decision represents one of the most prominent antitrust cases addressing self-preferencing in digital markets. The Commission found that Google systematically favored its own comparison-shopping service in general search results while demoting rival CSS. Several studies examine the competitive implications of this decision and Google's subsequent compliance measures. Hoppner (2022) argues that the remedy introduced by Google, allowing rival CSS to bid for placement in the Shopping Unit, did not address the main concern of the decision which was the preferential display of Google's own comparison service in general search results. Marsden (2020) also questions whether the new design effectively restores competition. The Product Listing Ads (the individual ad inside the Shopping Unit) often direct users directly to merchants rather than to comparison platforms, not increasing the visibility of the CSS. The case has also generated extensive legal debate regarding the interpretation of Article 102 TFEU and the treatment of self-preferencing as a form of abusive discrimination (Bostoen, 2022; Hornkohl, 2022; Lindeboom, 2022). The auction mechanism introduced after the decision has also been analyzed from a market design perspective. Frank et al. (2023) study Google's compliance mechanism as a position auction in which Google simultaneously acts as auctioneer and bidder. They show that this dual role can create incentives for strategic overbidding, increasing participation costs for rival CSS.

The Google Shopping case also contributed to the policy debate that led to the adoption of the Digital Markets Act (DMA), which introduces ex-ante obligations for large digital platforms designated as gatekeepers. The DMA explicitly prohibits certain forms of self-preferencing. A growing literature discusses the design and enforcement of this regulatory framework. Crémer and de Streel (2023) and Ibáñez Colomo (2023) analyze the institutional design of the DMA and its interaction with EU competition law, while Akman (2022) highlights potential concerns regarding legal certainty and regulatory flexibility. Other contributions emphasize the challenges of enforcing self-preferencing prohibitions in practice (Carugati, 2022; Franck and Peitz, 2024). There is also some recent work on the implementation of the DMA. Waldfogel (2024) finds that after the DMA designation of Amazon as a gatekeeper, the ranking advantage of Amazon's own products declined significantly. Closest to our setting, Püplichhuisen and Sirries (2026) document Google's 2024 redesign of its general search results page for product-related queries under Article 6(5): while a new dedicated element substantially increased the visibility of competing CSS, Google simultaneously introduced another product-comparison element, one that plausibly functions as a de facto CSS of its own, that is triggered more often and displayed more prominently, a pattern they interpret as a potential indicator of continued unequal treatment. In contrast, Pape and Rossi (2025) show that removing the direct link to Google Maps from search results increased searches for Google Maps but did not generate a significant increase in traffic to competing mapping services. Calzada et al. (2026) study the removal of the Google Flights module from search results following the DMA. The authors find that it increased organic traffic towards airline and flight comparison websites, with stronger effects for smaller platforms.

3 Institutional Background

3.1 The Commission's Decision and Google's Remedy

Between 2009 and 2015, the European Commission received numerous complaints against Google from rival CSS, online platforms, publishers and other digital market participants regarding Google's SERP. Several national competition authorities also transferred cases to the Commission, leading to the consolidation of the proceedings under a single case file (COMP/C-3/39.740 – Google Search). The complainants alleged that Google was abusing its position in general search by favoring its own specialized search service, Google Shopping, over competing comparison-shopping services. For the purposes of the competition assessment, the Commission defined two distinct relevant product markets: (i) general search services and (ii) CSS. General search services were considered a separate market, given their limited substitutability with other online services. CSS were also defined as a distinct market, as they enable users to compare offers across multiple online retailers and act as intermediaries redirecting traffic, rather than transactional destinations such as marketplaces.

The Commission's investigation identified two practices. The first involved prominently displaying Google Shopping, its own comparison-shopping service, at the top of search results using the visually appealing "One Boxes" feature (see Figure 17 for an illustration of this format). The second involved reducing the ranking of other comparison-shopping services in the general search results by using Google's algorithm to place them in lower positions. In particular, following Google's algorithm update Panda in 2011, introduced as a quality-improving update targeting low-value content, many CSS experienced a sharp and persistent decline in visibility. Empirical evidence covering the period 2010-2016 for the United Kingdom, Germany, France, Italy, and Spain is collected in the antitrust procedure. The evidence shows a clear and steady decline in the visibility and traffic in competing CSS following 2011. Consequently, the Commission concluded that: (1) the preferred position of Google's CSS on the general results page had a significant impact on its click-through rate; (2) these practices significantly decreased traffic to competing CSS; and (3) traffic to Google's own CSS increased as a result.

On 27 June 2017, the European Commission concluded that Google had abused its dominant position in general search by systematically favouring its own comparison shopping service (CSS) while demoting rival CSS in its search results, in violation of Article 102 TFEU. Given Google's very high market share in general search (typically exceeding 80% on desktop and 90% on mobile in most EEA countries during the analyzed period), general search traffic is considered an indispensable source of user acquisition for CSS; in the absence of a relevant alternative, Google's conduct was therefore able to reinforce its dominant position in the CSS market. The Commission imposed a fine of €2.42 billion and ordered Google to bring the conduct to an end within 90 days.[1]

In September 2017, Google did not remove the Shopping Unit from the search results page. Instead, it reconfigured the system by allowing third-party CSS to participate in the Shopping Unit through a bidding mechanism. Google stated that rival comparison services would be

[1]European Commission, "Antitrust: Commission fines Google €2.42 billion for abusing dominance as search engine by giving illegal advantage to own comparison shopping service," 27 June 2017. https://ec.europa.eu/commission/presscorner/detail/en/memo_17_1785.

treated on the same terms as its own service.[2] In its official documentation, Google states that Google Shopping operates as one CSS among others and that merchants can decide with which CSS they wish to work under equal conditions.[3] In practice, the remedy maintained the Shopping Unit while allowing third-party CSS to participate through the bidding mechanism. This design became the central point of debate regarding the effectiveness of the remedy.[4]

Figures 18 and 19 illustrate this change concretely. Before the remedy, the Shopping Unit displayed only product offers from Google's own comparison service, and rival CSS had no access to this prominent placement at all. After the remedy, third-party CSS can bid to have their product feeds included in the Shopping Unit alongside Google's own offers, in the form of Product Listing Ads (PLAs) – but their presence is limited to a small label (e.g. "by Kelkoo") beneath the product image, while the main clickable element continues to direct users straight to the merchant's website rather than to the CSS itself. Empirical click data collected by Hoppner (2020) indicate that around 99% of clicks within the Shopping Unit lead directly to merchants, and only about 1% correspond to links pointing to the participating CSS. In other words, the redesign didn't remove rival CSS from the Shopping Unit, but it didn't give them a real channel to attract traffic to their own sites either: they're now formally included, but they generate little traffic in practice. They gained a foothold in the product-feed pipeline, not a comparable way to attract users. We return to the competitive implications of this specific design choice in Section 7. Figure 20 in Appendix A illustrates this asymmetry directly with a live Shopping Unit listing.

Criticism from rivals emerged almost immediately. In February 2018, a group of 19 competing CSS wrote to Margrethe Vestager arguing that Google's proposed mechanism did not comply with the Commission's decision. According to these firms, allowing rival CSS to enter the Shopping Unit through auctions within a system controlled by Google did not restore genuine equal treatment.[5]

In November 2018, 14 European CSS sent another letter to Vestager raising similar concerns. The coalition expanded further in November 2019, when 41 CSS from 21 countries urged the Commission to intervene, arguing that Google continued to favour its own service and that the existing mechanism threatened the viability of competing comparison services.[6] [7] The main criticisms remained largely stable over time. First, rivals argued that the equal treatment offered by Google was formal rather than substantive, as the environment, rules, and visibility of the shopping unit continued to be designed by Google itself. Second, requir-

[2]Reuters, "Google to treat shopping rivals equally in line with EU order," September 2017. https://www.reuters.com/article/legal/google-to-treat-shopping-rivals-equally-in-line-with-eu-order-idUSKCN1C22B3.

[3]Google Merchant Center documentation, "CSS (CSS) program." https://support.google.com/merchants/answer/12652686.

[4]Reuters, "Google to treat shopping rivals equally in line with EU order," September 2017.

[5]Reuters, "Google's shopping rivals call for action from EU antitrust watchdog," February 2018. https://www.reuters.com/article/technology/googles-shopping-rivals-call-for-action-from-eu-antitrust-watchdog-idUSKCN1GC2RZ.

[6]Open letter from CSS to Commissioner Vestager, 2019. https://www.searchneutrality.org/google/comparison-shopping-services-open-letter-to-commissioner-vestager.

[7]Reuters, "Shopping rivals urge EU to act against Google for allegedly favouring own service," 2019. https://www.reuters.com/article/business/shopping-rivals-urge-eu-to-act-against-google-for-allegedly-favouring-own-servic-idUSL8N2875J7.

ing rivals to compete through bidding within Google's own advertising infrastructure merely displaced the competitive problem rather than resolving it. Third, Google Shopping continued to benefit from structural advantages derived from its integration within the Google Search ecosystem.

The European Commission did not adopt a second major non-compliance decision comparable to the original 2017 ruling. However, official sources indicate continued monitoring of Google's compliance. In its annual competition report published in April 2023, the Commission stated that it had continued to monitor the implementation of its decisions in both the Google Search (Shopping) and Google Android cases.[8] Public statements from the Commission also suggested ongoing concerns about the effectiveness of the remedy. In November 2019, Commissioner Vestager stated that Google's proposed solution "does not seem to be doing the trick," indicating that the mechanism did not appear to be helping rival CSS.[9] Overall, the Commission maintained a position of continued monitoring and pressure following the 2017 decision, without opening a new antitrust proceeding specifically focused on the compliance mechanism.

Google appealed the 2017 decision before the General Court of the European Union. On 10 November 2021, the General Court largely dismissed Google's appeal and upheld the €2.42 billion fine.[10] Google subsequently filed an appeal before the Court of Justice of the European Union in January 2022. In September 2024, the Court of Justice dismissed the appeal and definitively confirmed the Commission's decision and the fine.[11]

In summary, while the Commission has continued to monitor Google's compliance and has publicly expressed concerns about the effectiveness of the remedy, it has not issued a new formal decision regarding non-compliance. Between 2021 and 2024, European courts largely confirmed and then definitively upheld the original decision. As of 2025–2026, the CSS system remains in place and Google presents it as evidence of a competitive environment, although the effectiveness of the remedy continues to be debated.

[8]European Commission, "Annual Competition Report 2022." https://competition-policy.ec.europa.eu/system/files/2023-04/annual-competition-report_2022_report_part2_swd_en.pdf.

[9]Reuters, "EU's Vestager says Google's antitrust proposal not helping shopping rivals," November 2019. https://www.reuters.com/article/business/eus-vestager-says-googles-antitrust-proposal-not-helping-shopping-rivals-idUSKBN1XH2OY.

[10]Court of Justice of the European Union, Press Release No 197/21, "The General Court largely dismisses Google's action and upholds the €2.42 billion fine imposed by the Commission," 10 November 2021. https://curia.europa.eu/jcms/upload/docs/application/pdf/2021-11/cp210197en.pdf.

[11]Court of Justice of the European Union, Press Release No 135/24, 10 September 2024. https://curia.europa.eu/jcms/upload/docs/application/pdf/2024-09/cp240135en.pdf.



Figure 1: Timeline of the Google Shopping case (2017–2024).

3.2 Structure of Competition in the Comparison-Shopping Market

CSS (CSS) operate as two-sided platforms connecting consumers seeking product comparisons with merchants seeking visibility. CSS create value through indirect network effects: a larger consumer base attracts merchants, and more merchant offers improve comparison quality for users. Critically, CSS depend on search engine visibility. Most users reach CSS through search results, making visibility essential to their business model. This dependence on referral traffic from search engines is central to understanding how Google's preferential treatment of its own CSS affects competition in this market. In response to a user's query on Google Search, Google's general search results pages return search organic results, search text based advertisements, and results returned in "One Boxes" (for example, "Google Shopping", "Google Finance", "Google Flights", "Google Video"). These One Boxes present results in a visually prominent format and are integrated directly into the SERP. Among the different services, some operate on the basis of paid inclusion. In these cases, a third-party website must enter into an agreement with Google to be listed, typically under a pay-per-click system. Google Shopping operates under such a model, with its results displayed in a dedicated area of the SERP commonly referred to as the Shopping Unit (See Figure 17). The Shopping Unit comprises specialized search results from Google Shopping displayed in a visually prominent format within the general search results page. The individual product offers shown within this unit are commercially referred to as Product Listing Ads (PLAs). The PLAs operate under a cost-per-click (CPC) model, where merchants pay Google each time a user clicks the ad. These ads typically direct users straight to the merchant's website to complete a purchase.

CSS compete with one another primarily along three dimensions: the breadth of the merchant offers they gather, the comparison and filtering tools they provide to help users find the best price, and their ability to attract visitors through organic search and paid advertising. Because CSS do not sell products directly, their competitive success depends on directing as much search traffic as possible toward their own platform rather than toward specific merchants. For this reason visibility in search engines is determinant. The large majority

of the CSS in our sample are generalist comparators that compete for the same broad base of shoppers across most product categories, including Kelkoo, idealo, PriceRunner, Shopping.com, and Geizhals, differentiating themselves primarily through merchant coverage, user reviews, and price-alert features rather than through product specialization. A small number of CSS in our sample instead specialize in a specific product vertical. For example, PCPartPicker.com focuses exclusively on PC components, PriceWatch.com, centers on computer hardware and consumer electronics, and Shopalike.fr and ShopStyle.co.uk specialize in fashion, footwear, and related lifestyle categories. We retain these vertical CSS in our sample because they are comparison-shopping services to Google's visibility mechanism in the same way as the rest. Section 6.5 shows that our main results are robust to excluding them.

4 Data

Our dataset contains monthly web traffic information for CSS (CSS) collected from Semrush. The sample includes observations for five countries: France, Germany, the United Kingdom, the United States, and Canada. The European countries constitute the treated group affected by the Google Shopping decision, while the United States and Canada serve as control markets. The final dataset contains 3,200 monthly observations covering 77 unique CSS domains. Seven of these domains are observed in more than one country – for example, `shopping.com` is tracked separately in France, Germany, the United Kingdom, and the United States, and `nextag.com` in Canada, France, and the United States – because Semrush estimates traffic separately for each country-specific Google search market even when the underlying domain is the same. Counting each domain-country pair as a separate unit, consistent with the website-country fixed effects used in our empirical specification, the panel comprises 87 domain-country units: 24 in the United States, 20 in Germany, 18 in France, 17 in the United Kingdom, and 8 in Canada. Table 1 reports the corresponding distribution of monthly observations across countries. The largest share of observations comes from the United States (26.9%), followed by Germany, France, and the United Kingdom; Canada represents a smaller share of the sample. The time coverage of the dataset spans from January 15, 2016 to March 15, 2019 (39 monthly periods), allowing us to observe traffic patterns both before and after the European Commission's decision in June 2017. Table 2 presents summary statistics for the main variables used in the analysis. The dataset includes measures of organic visibility (organic keywords, organic traffic, and estimated organic traffic cost) as well as advertising activity (AdWords keywords, advertising traffic, and advertising cost).

Semrush estimates each of these variables from its own crawl of Google's organic search results and paid advertisements, using a proprietary combination of click-through-rate models and keyword search volumes rather than each website's own analytics.[12] *Organic keywords* counts the number of keywords for which a domain ranks among the top 100 organic (unpaid) results on Google. *Organic traffic* is the estimated number of monthly visits generated through these organic rankings, obtained by weighting each keyword's monthly search vol-

[12]Semrush, "Domain Overview," https://www.semrush.com/kb/254-domain-overview; Semrush, "Rank," https://www.semrush.com/kb/27-rank.

ume by the average click-through rate associated with the domain's ranking position. *Rank* is derived from the same estimated organic traffic: for each domain, Semrush multiplies the average click-through rate of its ranking positions by the monthly search volume of the corresponding keywords, summed across all of the domain's organic positions within Google's first ten result pages, to obtain a monthly organic traffic estimate. The domain with the highest estimated organic traffic receives Rank 1, so a *lower* value of Rank indicates *greater* organic visibility. Because Rank is constructed from this same estimated organic traffic figure, comparing each domain's estimate against all others' in the same month, we treat it as a complementary, relative measure of visibility rather than as independent evidence. *AdWords keywords* counts the number of keywords for which the domain's ads appear in paid search results. *AdWords traffic* is the estimated number of monthly visits generated through these paid results. *AdWords cost* is Semrush's estimate of the domain's actual monthly spend on the Google Ads keywords underlying these paid placements. From 2020 onward, Semrush also records the number of times each domain's offers appear within the Shopping Unit itself; this variable is not available for our main 2016–2019 sample period, but we use it in Section 7 to examine how CSS concentration within the Shopping Unit evolved in subsequent years.

The table reports statistics for the full sample as well as separately for treated and control markets. The dataset is an unbalanced panel because some CSS do not appear in all periods. Websites observed in fewer periods tend to be substantially smaller in terms of organic visibility and traffic. Appendix A Table 7 reports summary statistics by the number of periods in which a website appears in the dataset.

Because the dataset is an unbalanced panel, some CSS do not appear in all periods. In the baseline specification we restrict the sample to websites observed in at least 35 out of the 39 periods in the dataset. This restriction removes very small or transient websites with extremely low levels of traffic while preserving most of the variation in the sample (see Table 3). The remaining missing observations are limited and scattered across time. Websites observed in 35–38 periods typically miss only a small number of months. For example, ciao.com is missing four periods (event time 9, 10, 17, and 18), lowpriceshopper.co.uk is missing three periods (11, 12, and 17), and nextag.com is missing two periods (-17 and -16). These missing observations occur both before and after the treatment date and do not cluster around the intervention. This pattern suggests that the panel remains close to balanced and that the limited attrition is unlikely to bias the event-study estimates.

A separate, small set of website-country units is observed far more sparsely and is excluded from the baseline sample entirely. Four units fall in the 15–30 period range: buycheapr.com (United States, 16 periods), einfachbilliger.com (Germany, 26), pronto.com (United States, 15), and smarter.com (United States, 19). Three units are observed in fewer than 15 periods: smartshopping.com (Germany, 12), pricespy.ca (Canada, 5), and shoppydoo.fr (France, 5). As Table 3 shows, these seven units correspond to substantially smaller CSS: median monthly traffic is 17 for the 15–30 period group and 0 for the under-15 group, compared to 12,354 for websites with complete coverage. Their exclusion removes marginal, largely inactive players rather than economically relevant competitors in the comparison shopping market.

Table 1: Distribution of CSS by Country

Country	Observations	Percent
United States	861	26.91
Canada	274	8.56
Germany	740	23.12
France	666	20.81
United Kingdom	659	20.59
Total	3,200	100

Notes: The table reports the number of CSS (CSS) included in the dataset by country. European countries (France, Germany and the United Kingdom) constitute the treated markets affected by the Google Shopping decision, while the United States and Canada serve as control markets.

Table 2: Summary Statistics

	Control (US/CA)		Treated (EU)		Total	
Variable	Mean	Median	Mean	Median	Mean	Median
Rank	1,904,166	97,782	439,118	22,593	949,193	45,659
Organic keywords	327,826	21,169	130,666	33,078	199,310	29,128
Organic traffic	644,091	5,705	576,057	11,840	599,744	9,292
Organic cost	574,413	3,192	350,584	3,571	428,513	3,522
AdWords keywords	5,853	0	4,319	3	4,853	1
AdWords traffic	54,718	20	50,543	100	51,997	100
AdWords cost	59,659	0	25,228	0	37,215	0
Observations	1080		2022		3102	

Notes: The table reports summary statistics for the main variables used in the analysis. Organic variables correspond to traffic obtained from unpaid search results, while AdWords variables capture traffic generated through paid search advertising.

Table 3: Sample distribution

Coverage	**N obs**	**Median Rank**	**Median Traffic**
Complete (39 periods)	2,886	31,740	12,354
Near-complete (35–38)	216	502,914	217
Partial (15–30)	76	2,471,165	17
Marginal (<15)	22	11,016,094	0

Figure 2 plots the raw evolution of mean log organic traffic by country over the sample period, providing a first visual overview of the pattern examined more formally in the

event-study estimates of Section 6: organic traffic in the treated European countries and the control markets evolves broadly in parallel prior to the intervention, before diverging following Google's September 2017 redesign.

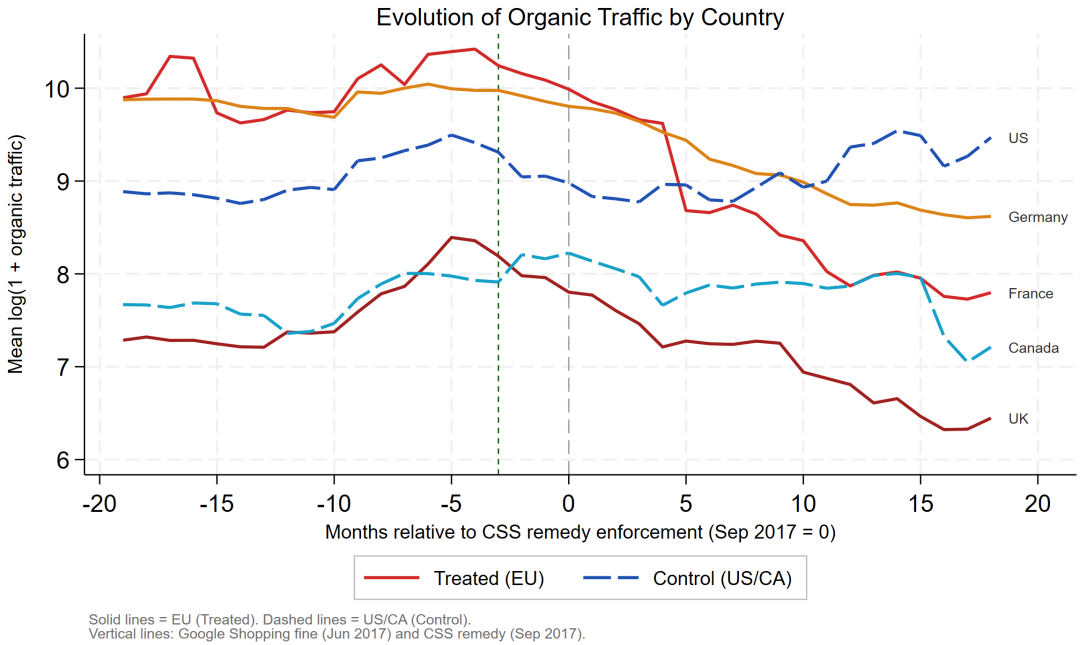


Figure 2: Evolution of mean log(1 + organic traffic) by country. Solid lines: treated European markets (France, Germany, United Kingdom). Dashed lines: control markets (United States, Canada). The dotted vertical line marks the Google Shopping fine (June 2017); the dashed vertical line marks the CSS remedy enforcement (September 2017).

5 Empirical Strategy

5.1 Identification

The objective of this paper is to evaluate the impact of Google's redesign of the SERP – introduced in September 2017 in response to the European Commission's Google Shopping decision – on the visibility of CSS in search results. The Commission's decision required Google to stop downgrading rival CSS in general organic search results, but it did not prescribe a specific remedy design. Google retained full discretion over how to comply and chose to reconfigure the Shopping Unit, allowing third-party CSS to participate in its advertising system through a bidding mechanism. Our empirical strategy therefore identifies the effect of this specific design choice, rather than the effect of the Commission's decision considered in the abstract.

We treat the implementation of Google's redesign as an exogenous shock affecting the visibility of CSS in Google Search. Google introduced this redesign only within the European Economic Area (EEA), while the SERP in non-European markets remained unchanged. This

geographic variation in where the redesign was implemented – not in the underlying legal decision itself, which applied only to Google's EEA conduct – allows us to identify the causal effect of the redesign by comparing outcomes for webpages in European countries (treated group) with webpages in non-European markets (control group).

Changes in the design of Google's search results may influence web traffic through several channels. To capture these mechanisms, we examine outcomes related to organic traffic, paid search traffic, and search ranking positions. Organic traffic measures visits generated through unpaid search results and is expected to be the outcome most directly affected by changes in search ranking or visibility. Because users disproportionately click on top-ranked results, even small changes in ranking positions can translate into large changes in organic traffic. The intervention may also affect firms' advertising strategies. In particular, the compliance mechanism introduced after the decision required CSS to participate in Google's Shopping advertising system in order to obtain visibility within the Shopping unit. As a result, firms may have increased their reliance on paid search advertising to maintain visibility. To account for this mechanism, we also analyze changes in paid search traffic. Finally, we examine changes in search ranking positions, which provide a direct measure of the visibility of webpages in Google Search results. This allows us to assess whether traffic effects are driven by shifts in the relative position of comparison shopping websites within the SERP.

5.2 Event-study Difference-in-Differences Estimates

To assess the dynamic effects of the intervention and evaluate the validity of the parallel trends assumption, we estimate an event-study specification that allows the treatment effect to vary over time before and after the implementation of Google's compliance mechanism.

Formally, we estimate the following specification:

$$\ln(Y_{ict}) = \sum_{j \neq -1} \beta_j (EU_c \times 1[t = j]) + \alpha_i + \gamma_t + \varepsilon_{ict} \tag{1}$$

where Y_{ict} denotes the outcome variable for website i in country c at time t; because a small number of CSS operate in more than one of our five countries (e.g. `shopping.com`), i indexes website-country pairs rather than domains alone, so each country-specific instance of a multi-country CSS is treated as a separate unit. The dependent variable varies across specifications and includes organic traffic, paid search traffic, and ranking measures. EU_c is an indicator equal to one for European countries affected by the Google Shopping decision and zero for non-European countries. The indicator $1[t = j]$ represents event time j relative to the implementation of Google's compliance mechanism. The omitted period corresponds to the period immediately before the intervention.

The coefficients of interest, β_j, capture the dynamic effect of Google's redesign at event time j relative to the pre-treatment baseline. Website-country fixed effects (α_i) control for time-invariant characteristics of each website-country unit, such as brand recognition, domain authority, or business model. Time fixed effects (γ_t) capture aggregate shocks affecting all websites simultaneously, including seasonality and changes in overall search demand. The event-study specification allows us to examine the evolution of treatment effects over time

and provides a graphical assessment of whether treated and control markets followed similar trends prior to the intervention.

5.3 Aggregate Difference-in-Differences Estimates

In addition to the event-study specification, we estimate an aggregate difference-in-differences specification in order to summarize the overall impact of the intervention in a single, easily reportable coefficient. Pooling all post-treatment periods into one estimate, rather than reporting 18 separate event-time coefficients, also increases statistical precision and facilitates comparison of the average effect size across the three outcomes. Formally, we estimate the following specification:

$$\ln(Y_{ict}) = \beta(EU_c \times Post_t) + \alpha_i + \gamma_t + \varepsilon_{ict} \tag{2}$$

where Y_{ict} denotes the outcome for website i in country c at time t. The variable EU_c is an indicator equal to one for European countries affected by the Google Shopping decision, while $Post_t$ equals one for periods after the implementation of Google's compliance mechanism.

The interaction term $EU_c \times Post_t$ identifies the treatment effect by comparing the change in outcomes for CSS in European markets before and after the intervention with the corresponding change observed in non-European markets over the same period. In other words, the coefficient β captures the difference between the pre–post change in treated markets and the pre–post change in control markets.

This specification therefore provides a summary measure of the average impact of Google's redesign on the outcomes considered, after accounting for time-invariant differences across websites through website-country fixed effects (α_i) and common time shocks through time fixed effects (γ_t).

The identifying assumption underlying this difference-in-differences strategy is that, in the absence of the intervention, CSS in European and non-European markets would have followed parallel trends in the outcomes considered. We assess the plausibility of this assumption using the event-study estimates presented in the following section. If treated and control webpages followed similar trends prior to the intervention, the pre-treatment coefficients should be statistically indistinguishable from zero.

5.4 Comparability of Treated and Control Markets

The choice of the United States and Canada as control markets reflects their status as large, high-income economies with mature CSS industries, in which Google was not required to alter its Search results following the 2017 decision. Several features support their comparability to the treated European markets specifically at the level of the CSS market itself, rather than of e-commerce in general. First, the same underlying business model is present in every country in our sample: all units in the panel are comparison shopping websites competing for referral traffic from Google Search. Table 2 shows that treated and control markets nonetheless differ in the levels of CSS activity (e.g., median organic and paid traffic are higher in the treated group); we return below to why such level differences do not threaten identification. Moreover, several major CSS operate as the same company across both treated and control markets – `shopping.com`, `nextag.com`, `bizrate.com`, `pcpartpicker.com`, `pricegrabber.com`,

and `shopzilla.com` are all observed in the United States and/or Canada as well as in one or more European countries – providing a natural within-firm benchmark that holds the underlying platform, technology, and business model fixed across the treatment and control groups. The remaining CSS, while distinct companies, operate the same type of business in every country of the sample; we do not have country-level data on CSS or Google Shopping usage specifically, so this within-sample evidence is the most direct comparison available.

Second, on the broader search and retail environment in which these CSS operate, the control markets fall within the same broad range as the treated markets rather than representing a categorically different environment. Google's average desktop search market share between January 2016 and December 2017 was approximately 80% in the United States and 87% in Canada, compared to 85%, 88%, and 90% in the United Kingdom, Germany, and France, respectively.[13] As general background on the retail environment – not a direct measure of CSS or Google Shopping usage, for which comparable country-level statistics are not available – e-commerce accounted for 8.0% of US retail sales in 2016 and 8.9% in 2017,[14] compared to an estimated 10% in France and 11% in Germany, while the United Kingdom – itself a treated market – stood out with a considerably higher share of approximately 23%.[15] This pattern shows that neither the treated nor the control group is internally homogeneous on these dimensions: the United States is somewhat less Google-dependent than the average treated country, but not categorically different from it, and the variation in e-commerce penetration across treated countries themselves (e.g., between the United Kingdom and France) is at least as large as the gap between the treated and control groups.

Ultimately, what the difference-in-differences design requires is not that treated and control markets be identical in levels, but that they would have followed parallel trends in CSS traffic absent Google's redesign. Website-country fixed effects absorb any time-invariant differences in levels across countries, including the market-structure differences documented above. We assess the parallel-trends assumption directly using the event-study estimates in Section 6: the pre-treatment coefficients are close to zero and statistically insignificant for all three outcomes (Figures 3, 4, and 5), providing the primary empirical support for the validity of the United States and Canada as control markets.

As a further, entirely independent check, Section 6.4 exploits within-country variation instead of cross-country variation: motivated by the heterogeneous effects documented in Section 6.3, we use large CSS as an implicit comparison group for small CSS within each set of countries, and show that the resulting gap opens up only in the treated European countries, not in a placebo version of the same test run within the United States and Canada.

6 Main Results

Throughout this section, we interpret the estimated coefficients as the causal effect of Google's September 2017 SERP redesign – the specific compliance mechanism Google chose

[13] Authors' calculation from StatCounter Global Stats monthly desktop search engine market share data, January 2016–December 2017, https://gs.statcounter.com/search-engine-market-share/desktop.

[14] US Census Bureau, Quarterly Retail E-Commerce Sales reports, 2016–2017. https://www.census.gov/retail/ecommerce.html.

[15] Statista, e-commerce share of retail sales in European countries, 2017.

to adopt – rather than as a direct test of the Commission's decision itself. This distinction matters for interpretation: a null or adverse result indicates that the design Google implemented did not restore competition, not that the underlying prohibition of self-preferencing was misguided.

6.1 Event-Study Results

Figures 3, 4, and 5 present the event-study estimates for organic traffic, ranking positions, and paid search traffic, respectively. In each case, the coefficients represent the difference between European and non-European markets relative to the period immediately before the intervention; for ranking, higher values correspond to worse positions in search results. Before the intervention, the estimated coefficients for all three outcomes remain close to zero and are not statistically significant, indicating that treated and control markets followed similar trends in the pre-treatment period and supporting the validity of the parallel trends assumption.

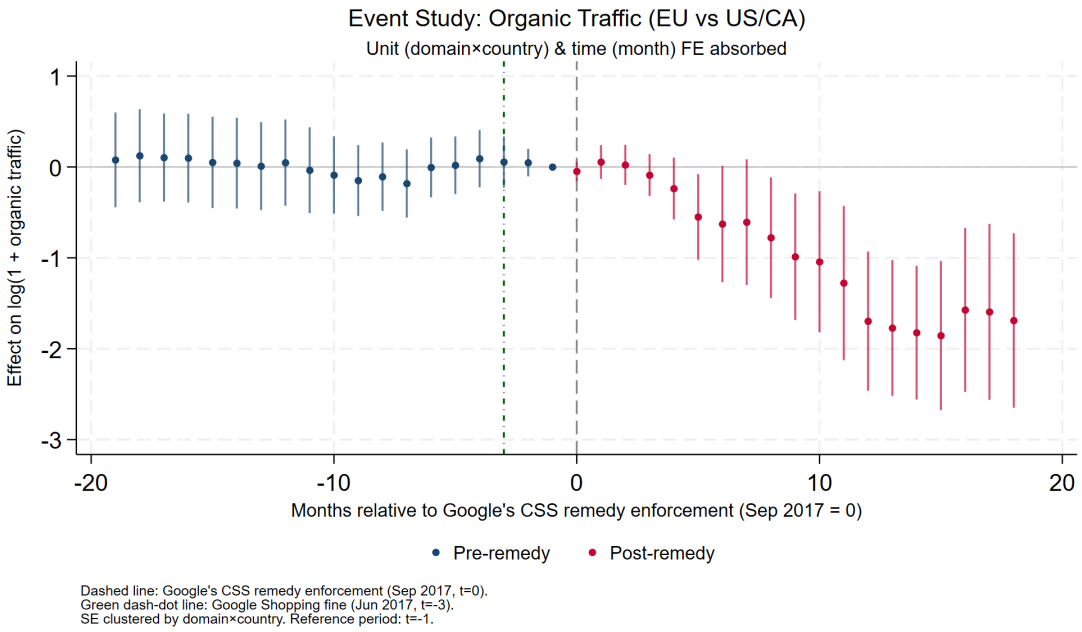


Figure 3: Event-study estimates of the effect of Google's redesign on organic traffic. The vertical dashed line indicates the implementation of Google's redesign.

Following the intervention, organic traffic shows no immediate shift; instead, the coefficients decline progressively over time (Figure 3). This pattern contrasts with the objective underlying the European Commission's decision, which aimed to prevent Google from systematically downgrading rival CSS in search results, and indicates that the specific redesign Google implemented did not achieve that objective. One possible explanation is that the compliance mechanism implemented by Google did not effectively restore the visibility of rival CSS in organic search results. This interpretation is consistent with existing discussions

in the literature on the Google Shopping case: Hoppner (2020) provides descriptive evidence from 21 European countries showing a continued decline in organic traffic to CSS after the adoption of the new mechanism, although that evidence is descriptive and does not rely on a causal identification strategy.

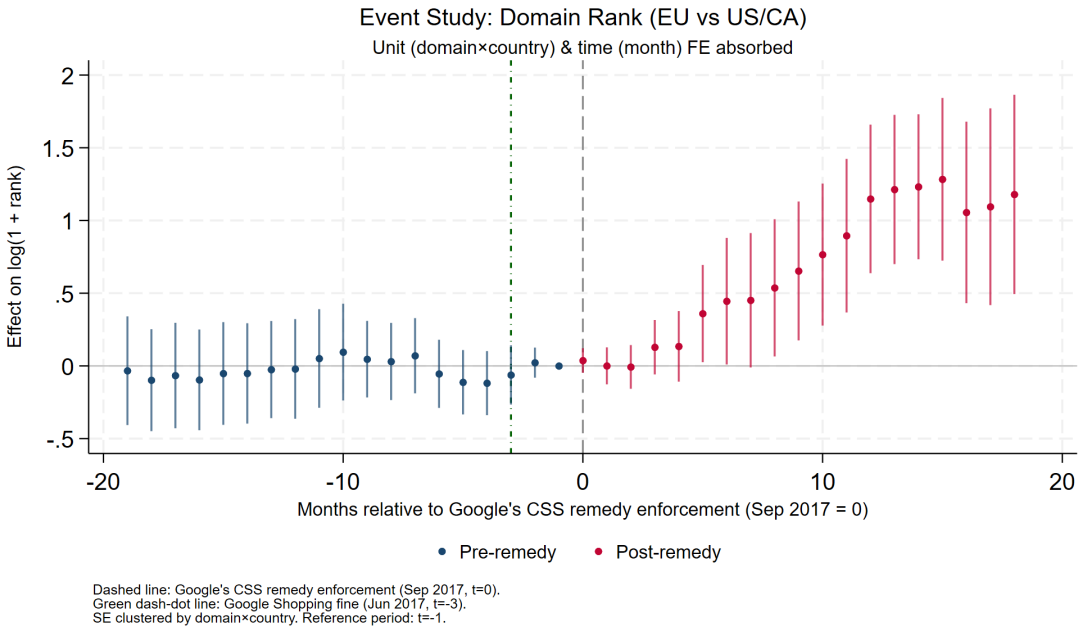


Figure 4: Event-study estimates of the effect of Google's redesign on ranking positions. The vertical dashed line indicates the implementation of Google's redesign.

Ranking positions follow a similar pattern (Figure 4): following the intervention, the coefficients become positive and increase over time, indicating that comparison shopping websites in Europe experienced a deterioration in their ranking relative to control markets. Because Rank is calculated from the same underlying organic traffic estimate discussed in Section 4, rather than from a domain's actual keyword-level position in Google's SERPs, this result is best read as confirming the organic traffic finding in a more interpretable, ordinal form, capturing a deterioration in CSS's relative organic visibility, rather than as fully independent evidence that specific CSS were moved to lower positions within individual keyword-level search results.

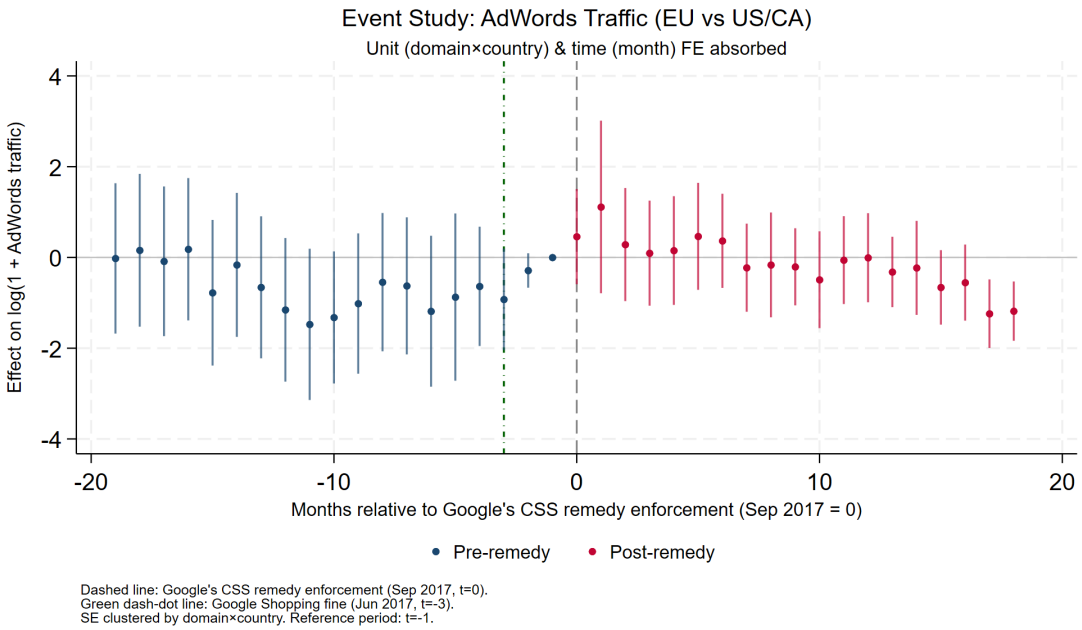


Figure 5: Event-study estimates of the effect of Google's redesign on paid search traffic. The vertical dashed line indicates the implementation of Google's redesign.

Paid search traffic, by contrast, shows no such divergence (Figure 5). Under Google's compliance mechanism, CSS can obtain visibility in the Shopping Unit by bidding for placement, a channel to which they had no access prior to the redesign. However, since CSS may already have been bidding on relevant search keywords through standard text ads before the redesign, it is not obvious how much additional paid traffic access to the Shopping Unit alone should generate, particularly since our AdWords traffic measure captures traffic reaching each CSS's own domain. As documented in Section 3, the vast majority of clicks within the Shopping Unit route directly to the merchant rather than to the participating CSS. Consequently, even absent any reallocation of advertising budgets across paid formats, gaining access to the Shopping Unit would be expected to generate little additional traffic to the CSS's own site, since most of the resulting clicks bypass it entirely. We therefore interpret the absence of a significant increase in paid traffic as consistent with this click-routing mechanism. In Section 7, we discuss several mechanisms that may account for this result.

6.2 Average Treatment Effects

While the event-study estimates above are useful for showing the dynamic evolution of the treatment effect, the aggregate difference-in-differences estimates are central for assessing its overall magnitude and statistical significance. Table 4 reports these estimates, corresponding to Equation (2) and summarizing the average post-treatment effect of Google's redesign across the three outcomes considered.

Table 4: Aggregate Difference-in-Differences Estimates

	(1) Organic Traffic	(2) Rank	(3) Adwords Traffic
EU × Pre	-0.0248	-0.0027	-0.6436
	(0.1436)	(0.1017)	(0.7061)
EU × Post	-0.9907***	0.6858***	-0.1279
	(0.2691)	(0.1872)	(0.3705)
Website-Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	2,886	2,886	2,886

Notes: The table reports aggregate difference-in-differences estimates corresponding to Equation (2). EU × Pre tests for pre-treatment differences between European and non-European markets, while EU × Post captures the average treatment effect after the implementation of Google's compliance mechanism. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The EU × Pre coefficients, which test for pre-existing differences between treated and control markets, are small and statistically insignificant for all three outcomes, corroborating the parallel-trends evidence from the event-study estimates above.

The EU × Post coefficients indicate a significant decline in organic traffic and a significant deterioration in ranking positions for CSS in European markets following the implementation of Google's compliance mechanism, while no statistically significant change is observed in paid search traffic. Because these coefficients are large, we translate them into percentage effects using the exact formula $100 \times (e^{\beta} - 1)$ rather than the linear approximation $100 \times \beta$, which is only reliable for small coefficients. The organic traffic coefficient of -0.99 corresponds to an average decline of approximately 63% relative to control markets. The ranking coefficient of 0.69 corresponds to an average deterioration of approximately 99% in the Rank metric, where higher values indicate worse visibility; this large percentage partly reflects the heavy-tailed nature of the Rank variable, which orders CSS among tens of millions of domains worldwide, and is consistent with the event-study estimates in Appendix A, where the corresponding coefficient grows well beyond this post-period average by the end of the sample.

6.3 Heterogeneous Effects Across CSS Websites

The impact of Google's redesign may differ across CSS depending on their pre-existing market position. In particular, larger platforms with higher pre-treatment traffic may be able to maintain a better position themselves in the search results and gather more organic traffic even with Google's behavior. Moreover, they also may be able to better participate in Google's advertising ecosystem, while smaller platforms may struggle to do so. To explore this possibility, we estimate a triple-difference (DDD) specification that allows the treatment effect to vary across groups of CSS defined by their pre-treatment traffic levels. We classify websites according to whether their pre-treatment traffic lies above or below a given percentile

threshold (median and 75th percentile). This classification is based exclusively on the pre-treatment period to ensure that group assignment is not affected by the intervention itself.

The estimation augments the event-study specification by interacting the treatment indicators with the group membership variable. Formally, the specification can be written as:

$$\ln(Y_{ict}) = \sum_{j \neq -1} \beta_j (EU_c \times 1[t = j]) + \sum_{j \neq -1} \gamma_j (EU_c \times High_i \times 1[t = j]) + \alpha_i + \gamma_t + \varepsilon_{ict} \quad (3)$$

where $High_i$ is an indicator equal to one for CSS whose pre-treatment traffic lies above the chosen percentile threshold. In this specification, β_j captures the treatment effect for websites below the threshold, while γ_j measures the differential effect for websites above the threshold. Figure 12 presents the results using the median cutoff, Figure 13 reports the estimates using the 75th percentile threshold, and Figure 14 reports the estimates using the 90th percentile threshold. Using the median cutoff, we do not find significant differences between large and small CSS. However, when we use the 75th percentile threshold, we find that larger CSS websites experience a smaller negative impact. This pattern is observed both in organic traffic and in ranking positions. At the 90th percentile threshold the pattern is, if anything, more pronounced: organic traffic and ranking positions deteriorate sharply and persistently for the bottom 90% of CSS, while the top decile shows no clear, sustained divergence from zero, indicating that the decline documented in the main results is concentrated among all but the very largest CSS. For paid traffic, the interpretation of the estimates is limited across all three thresholds due to violations of the parallel trends assumption.

Overall, these findings suggest that the redesign did not restore competitive conditions in the CSS market. While larger CSS appear to resist more the decline in organic visibility, the lack of improvement in paid traffic indicates that participation in the Shopping Unit did not compensate the loss of organic traffic. This asymmetry was anticipated by contemporaneous industry observers: writing in December 2018, shortly after Google discontinued SpendMatch – a cash-back incentive it had offered to merchants advertising through rival CSS – Searchmetrics predicted that CSS size would become "a decisive factor determining how competitive an external competitor CSS is able to be" under the new bidding system, as larger CSS managing more merchant budget could operate on thinner margins than smaller rivals.[16] This heterogeneity may also partly reflect name prominence rather than scale alone: Baye et al. (2016) show that failure to account for the prominence of a retailer's name, in addition to its position in search results, significantly inflates the estimated impact of screen position on organic clicks, since more recognizable brands retain organic traffic largely independently of ranking mechanics. Because larger CSS in our sample are plausibly also more established and recognizable brands, our size-based heterogeneity results may partly capture this name-prominence channel alongside the scale-related mechanisms discussed above.

[16] Search Engine Land, "Report: Google PLA antitrust remedy turning comparison engines into agencies," 10 December 2018, citing Searchmetrics' 2018 Shopping Study. https://searchengineland.com/report-google-pla-antitrust-remedy-turning-comparison-engines-into-agencies-309181.

6.4 Validation Using Within-Country Variation: Small vs. Large CSS

The heterogeneity results above, showing that larger CSS resist the decline in organic visibility better than smaller CSS, suggest a further way to validate our identification strategy that does not rely on cross-country comparability at all. If Google's redesign affects smaller CSS to a greater extent than larger ones, this size gradient should open up specifically *within* the treated European countries after September 2017, and should not appear in an equivalent within-country comparison in the United States and Canada, where no redesign occurred.

We implement this test by reusing the same classification as in Section 6.3, now computed within each group of countries rather than pooling the full sample: CSS with pre-treatment organic traffic above the 75th percentile, and, as a robustness check, above the stricter 90th percentile, are classified as large, with the remainder classified as small. We then estimate an event-study specification with small-CSS-by-event-time interactions, website-country and month fixed effects, and standard errors clustered by website-country, separately for the treated European countries and for the untreated United States/Canada sample, for each of our three main outcomes: organic traffic, ranking position, and paid (AdWords) traffic.

Figure 15 in Appendix A presents the results within the treated European countries, using large CSS as a comparison group. For organic traffic, pre-treatment coefficients are broadly stable, though somewhat noisier than in our main cross-country design; following the redesign, the gap between small and large CSS widens sharply and persistently, matching the pattern of our main organic traffic results in Section 6. Ranking positions move in the same direction: since higher values indicate worse rankings, the corresponding gap widens after the redesign as well. Both patterns are essentially unchanged when CSS are instead classified using the stricter 90th percentile cutoff. Paid traffic, by contrast, shows confidence intervals that remain wide and overlap zero both before and after the redesign under either cutoff, consistent with the parallel-trends concerns already noted for this outcome in Section 6.3.

Figure 16 in Appendix A repeats the identical exercise within the United States and Canada. If the growing gap between small and large platforms documented above were part of a broader trend in the CSS industry rather than being caused by Google's redesign specifically, we would expect to see a similar divergence show up here too. Instead, for both organic traffic and ranking positions, the estimated gap remains close to zero and statistically insignificant throughout the post-period, at both the 75th and 90th percentile cutoffs, consistent with the absence of any equivalent redesign in these markets. Paid traffic remains noisy and centered near zero in this placebo sample as well.

Together, these exercises show that the pattern documented in our main results – smaller CSS losing disproportionately more visibility than larger CSS, in both organic traffic and ranking positions – emerges specifically where and when Google's redesign was implemented, and is absent from an identical within-group comparison in an unaffected market. This provides additional support for our identification strategy that is independent of the cross-country comparability discussed in Section 5.4.

6.5 Robustness Checks

As a further robustness check, we re-estimate our main specification excluding the four CSS in our sample that specialize in a narrow product vertical rather than competing as general-purpose comparators (PCPartPicker.com, PriceWatch.com, Shopalike.fr, and ShopStyle.co.uk; see Section 3). Figures 9, 10, and 11 in Appendix A show that excluding these four vertical CSS leaves the estimated effects on organic traffic, ranking positions, and paid search traffic virtually unchanged, indicating that our main results are not driven by CSS operating in a different competitive segment from the rest of the panel.

As a robustness check, we re-estimate the event-study specification using a balanced panel of CSS websites. In this specification, we restrict the sample to websites that are observed throughout the entire time window and estimate the effects over a window of 39 periods. The resulting estimates are presented in Appendix A. Figures 6, 7, and 8 show patterns that are highly consistent with the baseline results. The evolution of organic traffic, ranking positions, and paid search traffic follows the same evolution observed in the main specification, indicating that the results are not driven by changes in the composition of the sample over time.

As a further robustness check, we address the concern that our two-way fixed-effects estimates implicitly weight the two control countries equally, which may be problematic if pre-treatment trends differ across markets and the number of treated and control countries is small. We re-estimate the aggregate treatment effect using the Synthetic Difference-in-Differences (SDID) estimator of Arkhangelsky et al. (2021), which reweights control countries and pre-treatment periods to better match the treated group's pre-trend, nesting both two-way-FE DiD and synthetic control as special cases. Because SDID operates on a small number of aggregate units rather than on our full domain-country panel, we first collapse the data to a country-month panel (5 countries $\times$ 38 months). Table 5 compares the resulting SDID estimates to two-way FE DiD estimated on this same collapsed panel, and to our main-text estimates from the domain-country panel. Because placebo (permutation) inference requires strictly more control than treated countries – a condition our sample does not satisfy with only two control countries – we use jackknife standard errors, the standard fallback recommended by Arkhangelsky et al. (2021) in this situation.

Table 5: Robustness: Synthetic Difference-in-Differences

	Organic Traffic	Rank	AdWords Traffic
Main-text DiD (domain-country panel)	-0.991***	0.686***	-0.128
	(0.269)	(0.187)	(0.371)
Two-way FE DiD (country panel)	-0.985**	0.693**	0.249
	(0.257)	(0.184)	(0.546)
Synthetic DiD (country panel)	-0.981***	0.693***	0.503
	(0.272)	(0.197)	(0.618)
Observations (domain-country panel)	2,886	2,886	2,886
Observations (country panel)	190	190	190

Notes: The main-text DiD row reproduces the EU × Post coefficients from Table 4, estimated on the domain-country panel with website-country and time fixed effects. The remaining two rows are estimated on a country-month panel obtained by collapsing the domain-level data (winsorized at the 1st/99th percentiles) to country means; the two-way FE row applies the same fixed-effects specification to this coarser panel, isolating the effect of aggregation from the effect of reweighting. Synthetic DiD standard errors are jackknife-based. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Organic traffic and ranking estimates are remarkably stable across all three specifications, both in magnitude and statistical significance, indicating that our main results are not an artifact of implicitly equal-weighting the two control countries. The AdWords traffic coefficient remains statistically insignificant throughout, consistent with our conclusion that the redesign did not generate a significant paid-traffic response; its point estimate is more sensitive to the choice of specification, but given that none of the three estimates are distinguishable from zero, this sensitivity does not affect the paper's conclusions.

6.6 Ruling Out Demand-Side Confounds

A potential concern is that the decline in organic traffic to CSS reflects a contraction in underlying search demand, rather than the effect of Google's SERP redesign. We present three pieces of evidence against this alternative explanation.

First, to assess whether overall search demand followed differential trends across treated and control markets, we collect Google Trends data for more than 50 product categories representative of CSS activity. Figure 21 shows that search interest remained stable across both European and non-European markets throughout the study period, with no divergence around either the EC fine (June 2017) or the redesign implementation (September 2017). Second, Google's search engine market share in Europe remained broadly constant at approximately 90% throughout the study period, with no significant change around either the EC fine (June 2017) or the redesign implementation (September 2017), ruling out substitution toward alternative search engines as a confounding factor (StatCounter Global Stats, 2017). Third, the share of individuals using search engines in treated countries grew continuously over the same period: from 85.6% to 89.4% in France, from 89.9% to 92.6% in Germany, and from 94.9% to 95.6% in the United Kingdom between 2016 and 2019 (Eurostat, 2024).

Taken together, these patterns confirm that aggregate demand flowing through Google

did not decline. The loss of organic traffic to rival CSS therefore reflects a within-SERP reallocation driven by Google's redesign of the interface, a mechanism we examine in detail in Section 7.

A natural follow-up question is where this reallocated traffic ultimately lands. Our data do not allow us to track individual users after they leave a CSS's organic listing, so we cannot directly identify the traffic's destination. However, the institutional evidence discussed in Section 3 is informative: within the Shopping Unit itself, the vast majority of clicks route directly to the merchant's website, since Google's own offers and those submitted by participating CSS are both PLA-style product listings that route clicks to the merchant rather than to whichever CSS supplied the underlying feed. This suggests that at least part of the reallocated traffic is captured by merchants directly through the Shopping Unit. We interpret this as suggestive rather than conclusive, since our data cannot rule out that some of the reallocated traffic also flows to other search results, general retailer websites reached independently of the Shopping Unit, or other channels outside our data. A related limitation is that our data do not allow us to fully disentangle whether the decline in organic traffic reflects a genuine deterioration in CSS's keyword-level ranking positions, a reduction in click-through rates at an unchanged ranking caused by the Shopping Unit crowding out attention to organic results, or some combination of both. This ambiguity does not affect our main conclusion, however: both channels are consistent with, and indeed anticipated by, the visibility displacement mechanism discussed in Section 7.

7 Mechanisms

The empirical results presented above raise a natural question: through what channels did Google's redesign fail to restore competition? Section 6.6 already ruled out an alternative, demand-side explanation for the observed decline. We now propose two complementary explanations, both concerning specific features of the design Google chose to implement in response to the Commission's decision. The first concerns the structure of the Shopping Unit itself; the second provides direct empirical support for the resulting displacement mechanism. We return, in the Conclusion (Section 8), to a related question: why Google has not voluntarily adopted an equivalent redesign in unregulated markets such as the United States.

7.1 Design of the Shopping Unit

A key limitation of Google's compliance mechanism lies in the design of the Shopping Unit itself. As described in Section 3, although CSS can technically participate in the system by submitting product feeds and bidding for placement, the structure of the interface continues to direct user attention primarily to merchants rather than to the comparison platforms. Consequently, even when CSS participate in the advertising system, the resulting traffic largely bypasses their platforms. Moreover, the new structure of the Shopping Unit may change user perception of competitive options. Before the redesign, users viewing the Shopping Unit encountered only Google's CSS, creating an incentive to scroll down to organic results to find alternative comparison services. After the redesign, the Shopping Unit displays multiple CSS options competing alongside each other, creating the perception of a

competitive marketplace within Google's space. This visible expansion of options within the Shopping Unit, from monopoly to a more competitive system, may reduce user incentives to scroll down and search for organic CSS results, since the number of alternatives increases and this greater competition may also introduce more competitive prices. CSS listings, for example, often display products at discounted prices. This visibility displacement effect is particularly damaging for CSS already positioned in lower organic rankings. While larger CSS with strong organic positions (first or second ranking) remain visible in the initial search results alongside the Shopping Unit, smaller CSS operating in lower ranking positions may lose organic traffic and user engagement, as users perceive their needs already met by the Shopping Unit options. This mechanism is consistent with a broader theoretical result on price-comparison platforms. Baye and Morgan (2001) show that adding more competing sellers to a price-comparison listing lowers the average price these sellers charge, a prediction confirmed empirically by Baye et al. (2004) using transaction data from an online price-comparison site.

As shown above, participation in the Shopping Unit does not restore meaningful traffic to CSS's own platforms. Even so, it can still generate profits for CSS acting as intermediaries within Google's advertising system. This apparent expansion of competition in the Shopping Unit should nonetheless be interpreted with caution, for two related reasons. First, much of it reflects participation by non-traditional actors rather than genuine comparison platforms. There are advertising agencies registered as CSS partners that do not operate as comparison services, but they manage merchant advertising campaigns. This distinction maps onto two economically different types of CSS. So-called reseller CSS, such as Channable or DataFeedWatch, manage a merchant's own advertising budget for a flat monthly fee and have no platform of their own to which they need to attract users. For these operators, the fact that PLA clicks route to the merchant rather than to the CSS is largely immaterial, since their revenue does not depend on retaining the user. Independent CSS such as Kelkoo, idealo, PriceRunner, and Foundem, by contrast, operate under the traditional comparison-shopping business model, in which revenue depends on directing users to their own platform, so a click that bypasses their site delivers them no benefit. This is consistent with Commissioner Vestager's own assessment, discussed in Section 3, that Google's mechanism was not delivering meaningful benefits to rival CSS. An industry report published in December 2018 shows how quickly this nominal participation grew. The share of PLA results occupied by CSS competitors rose from under 1% in the United Kingdom and about 2% in Germany in January 2018 to roughly a third of PLA results across the United Kingdom, France, and Germany combined by fall 2018. This growth, however, reflects how often CSS names appeared on the page, not how much traffic or how many clicks they actually received. The same report explained the discrepancy we mentioned. Many of the CSS filling these slots were not established price-comparison sites at all, but agencies managing merchants' bidding campaigns.[17] This pattern proved persistent rather than transitional. A 2022 study of over one million Shopping Units, submitted by a coalition of more than 40 CSS to Commissioner Vestager, found that just the top 20 CSS partners accounted for 93% of the product offers

[17]Search Engine Land, "Report: Google PLA antitrust remedy turning comparison engines into agencies," 10 December 2018, citing Searchmetrics' 2018 Shopping Study. https://searchengineland.com/report-google-pla-antitrust-remedy-turning-comparison-engines-into-agencies-309181.

displayed within Shopping Units appart from Google's CSS, yet the same top 20 partners generated only 1.4% of organic search results in the underlying dataset. This confirms, four years later, that the entities dominating the paid Shopping Unit were still primarily facilitating Google's advertising system rather than operating as genuine, independently-visible comparison platforms.[18]

Second, under this bidding system, sustained participation in the Shopping Unit appears to require scale, which may itself help explain the entry of agencies documented above. By aggregating many merchants' campaigns into a single bidding operation, an agency can reach a scale that an individual, independent CSS typically cannot, including the formal requirements for participation itself. Google's CSS program requires an applicant to display offers from at least 50 distinct merchant domains in each country where it operates, and to run its own product-search infrastructure. An operation managing many merchants' feeds clears this hurdle far more easily than a small, independent comparator.[19] It is mainly the large CSS that find it profitable to remain, while smaller ones struggle to do so. This becomes visible in our own data once we look specifically at CSS with a genuine, independently verifiable organic presence, as opposed to the broader, agency-dominated aggregate discussed above. Using Semrush's count of the number of times each domain's offers appear within the Shopping Unit – available from 2020 onward, as noted in Section 4 – we compute, for each of the three treated countries, the share of total Shopping Unit appearances captured by CSS classified according to their size. Table 6 shows that the top quartile already captured the large majority of Shopping Unit appearances in 2020, and that its share rose further through 2023 in all three countries – reaching 91% in France, 95% in Germany, and nearly 100% in the United Kingdom. Economies of scale plausibly reinforce this pattern. A CSS bidding on behalf of a larger aggregate merchant budget accumulates more conversion data, allowing its automated bidding to be calibrated more precisely and potentially achieving a better cost-per-click than a smaller rival with less data to learn from. While we are not aware of primary data documenting this channel directly, it would be consistent with Frank et al. (2023)'s finding that Google's dual role as auctioneer and bidder already raises the cost of participation for rival CSS.

[18] TechCrunch, "Google antitrust complainants call for EU to shutter its Shopping Ads Units," 18 October 2022, citing a joint letter from more than 40 rival CSS to Commissioner Margrethe Vestager. https://techcrunch.com/2022/10/18/eu-antitrust-complaint-google-shopping-units/.

[19] Google, "CSS program requirements," Comparison Shopping Services Help. https://support.google.com/css-center/answer/7524491?hl=en.

Table 6: Share of Shopping Unit appearances by CSS size, 2020–2023 (%)

Country	CSS size	2020	2021	2022	2023
France	Large (P75+)	72.7	81.9	87.9	91.2
	Medium (P50–P75)	23.0	15.4	11.3	8.7
	Small (<P50)	4.3	2.7	0.8	0.0
Germany	Large (P75+)	90.7	91.0	92.2	94.9
	Medium (P50–P75)	6.2	7.1	7.1	4.6
	Small (<P50)	3.1	1.9	0.7	0.5
United Kingdom	Large (P75+)	90.9	95.7	98.6	99.5
	Medium (P50–P75)	8.8	4.0	1.3	0.4
	Small (<P50)	0.4	0.3	0.1	0.1

Notes: CSS classified as large (top 25%), medium (50th–75th percentile), or small (below the 50th percentile) based on mean organic traffic.

Taken together, these features fundamentally change the way CSS could obtain visibility in Google's search results. Prior to the redesign, CSS relied primarily on organic search rankings to attract users; after the redesign, the Shopping Unit offers an alternative, paid visibility channel that, for genuinely independent CSS, delivers visibility of products without visibility to the CSS themselves. As a result, access to the Shopping Unit fails to compensate for the losses in organic visibility documented in our main results.

7.2 Empirical Evidence: SERP Features and Organic Traffic Displacement

To provide empirical evidence for the visibility displacement mechanism described above, we examine the relationship between the presence of prominent SERP features and organic traffic to CSS. While we lack information on the number of times the Shopping Unit appears for our main 2016–2019 sample period, we can use data on carousel-type SERP features as a proxy for understanding how top-of-page visual features affect organic traffic. Google's carousel is a horizontal row of visual search results such as products, local businesses, or reviews displayed without leaving the SERP.

Carousel features serve as an appropriate empirical analogue to the Shopping Unit. Both are visually prominent boxes positioned above organic search results that reduce the visible space dedicated to organic listings. By examining how carousel presence correlates with organic traffic to CSS, we can assess whether our proposed visibility displacement mechanism operates in practice. We estimate the following specification:

$$\log(\text{Organic Traffic})_{it} = \beta \log(\text{Carousel})_{it} + \gamma \log(\text{Rank})_{it} + \delta \log(\text{AdWords})_{it} + \alpha_i + \mu_t + \varepsilon_{it} \quad (4)$$

where $\log(\text{Carousel})_{it}$ indicates the presence and intensity of carousel features, $\log(\text{Rank})_{it}$ controls for organic ranking position, and $\log(\text{AdWords})_{it}$ captures paid advertising activity.

We include website-country fixed effects (α_i) and month fixed effects (μ_t) to account for time-invariant site characteristics and common time shocks. Table 9 in Appendix A presents the results. The coefficient on carousel is negative and statistically significant, indicating that the presence of prominent carousel features is associated with a 14% reduction in organic traffic to CSS. This finding is consistent with our theoretical mechanism: when Google displays prominent boxes (like carousels) at the top of search results, organic traffic to CSS websites declines, as users encounter these boxes before scrolling to organic results.

8 Conclusion and Policy Implications

The fundamental challenge in regulating self-preferencing lies in a structural paradox: regulators can prohibit discriminatory behavior, but platforms control how they comply. This creates an inherent tension between the regulator's ability to prohibit conduct and the way the platforms design the remedy. Thus, platforms can follow the rules on paper while undermining them. For example, through design choices (e.g., how the interface looks, how algorithms rank results, etc.). This paper uses the European Commission's 2017 Google Shopping decision as a laboratory to study this challenge. We examine the Google Shopping case because it combines three elements: a clear, documented infringement of competition law; an observable remedy whose implementation can be measured; and exogenous geographic variation (European intervention vs. no intervention in control markets) that allows causal identification. Yet our findings speak beyond Google. They reveal a problem for regulators in digital markets: how to design and enforce remedies when the regulated firm controls the mechanisms through which those remedies must operate.

This structural paradox is reinforced by evidence on Google's incentives. If the EU compliance mechanism reduces rival CSS traffic while requiring Google to share Shopping advertising revenue with competing CSS platforms, why has Google not voluntarily extended it to unregulated markets such as the United States? In the US, Google retains all Shopping advertising revenue; under the EU mechanism, participating CSS platforms take a cut of that revenue, and the mechanism itself forces Google to act simultaneously as auctioneer and bidder, creating incentives for strategic overbidding that reduce auction efficiency (Frank et al., 2023). Absent regulatory pressure, Google has no incentive to adopt a design that reduces its own revenue, even though that same design hurts competitors more than it hurts Google – consistent with Google implementing the mechanism only where legally required to do so. The financial evidence is consistent with this account: according to Google's accounting data, advertising revenue in the United States grew 31% between 2017 and 2019, compared to slower growth in the regulated United Kingdom (18%) and France (23%). This asymmetry underscores why remedies that leave design discretion to the regulated firm are unlikely to be adopted, let alone extended, absent sustained regulatory pressure.

Our results show three main findings. First, organic traffic to CSS in Europe declined further relative to non-European control markets after the redesign's implementation. Second, the redesign did not generate significant increases in paid search traffic. Third, larger CSS platforms better resisted organic traffic decline, yet even they experienced no improvements in paid traffic. These patterns suggest the redesign did not restore competition in the CSS market.

Rather than improving rival CSS visibility, Google's compliance mechanism changed how CSS could obtain visibility in the SERP, yet those changes failed to benefit these platforms. Rival CSS also experienced continued relative deterioration in organic ranking positions, a pattern consistent with continued downgrading in organic search, whether through algorithmic demotion or through the reduced organic visibility documented in Section 6.6. At the same time, Google created a paid advertising system in which CSS appear as product feed intermediaries rather than as independent comparison platforms, and this paid alternative does not appear to restore competition in the CSS market. Additionally, the shift from organic to paid traffic under the adopted mechanism is consistent with a new competitive structure that disadvantages smaller CSS. Decarolis and Rovigatti (2021) show how concentrated intermediaries can exercise power in auctions, coordinating bids and segmenting markets to reduce costs at the expense of smaller competitors; our finding that larger CSS better resist the decline in organic visibility, without commensurate gains in paid traffic even for them, is consistent with a similar dynamic, in which requiring CSS participation through Google's advertising system favors larger, well-capitalized competitors capable of more sophisticated bidding strategies and data pooling, though we do not observe bidding behavior directly. The redesign is therefore best described as having created a system that disadvantages smaller competitors, rather than one that straightforwardly restores their competitive position.

Beyond its effects on competing CSS, this new intermediation structure may also carry consequences for final consumers. Prior to the redesign, a merchant advertising through the Shopping Unit faced a single layer of markup, Google's own cost-per-click price. Under the current mechanism, a merchant that instead advertises through a CSS intermediary faces two independently set markups: Google's price for Shopping Unit placement, and the CSS's own fee or margin for managing the bid on the merchant's behalf. This structure raises the possibility of double marginalization (Spengler, 1950). Because each layer sets its price without internalizing the effect on the other's profit-maximizing margin, the resulting advertising cost to merchants can exceed what a single vertically integrated price-setter would charge, a cost that may ultimately be passed through, at least in part, to the retail prices consumers pay. Notably, this logic applies asymmetrically across participating CSS despite the formally identical bidding mechanism Google requires of all of them. Google states that its own comparison service operates as "one CSS among others," subject to the same procedure as independent rivals. Yet because Google Shopping and Google Ads remain under common ownership, Google's own channel need not face the independent margin-setting that Spengler's logic implies, since a common owner can set a single, joint profit-maximizing price across both layers rather than two uncoordinated ones (for a related treatment of double marginalization under platform first-party selling, see Dendorfer, 2024). The resulting cost disadvantage should therefore fall disproportionately on merchants routing through independently-owned CSS rather than through Google's own comparison service, compounding the traffic-based disadvantage documented above with a potential cost-based one. We are unable to test this channel directly, as we observe neither the prices of products advertised within the Shopping Unit over our sample period nor Google's internal transfer pricing between its own comparison and advertising functions; we flag it here as a plausible additional welfare channel worth investigating once such data become available.

These insights are particularly important for the Digital Markets Act and emerging reg-

ulatory frameworks globally. The DMA introduced a considerably stronger legal instrument than the abuse-of-dominance framework underlying the 2017 decision: Article 6(5) explicitly prohibits gatekeepers from treating their own services more favourably than third-party services in ranking, an ex-ante obligation backed by the threat of periodic penalty payments of up to 5% of worldwide daily turnover for continued non-compliance. Whether this stronger instrument succeeds where the 2017 remedy failed is no longer a purely hypothetical question. On 23 July 2026, nearly a decade after the original Google Shopping decision, the Commission found Google non-compliant with this obligation and imposed a €460 million fine for self-preferencing its own shopping, hotels, transport, and sports results in Search rankings, alongside a separate €430 million fine for anti-steering violations on Google Play – a combined €890 million penalty, the first major DMA fine issued to date.[20] The same month brought a second, independent blow: on 1 July 2026, the Stockholm Patent and Market Court ordered Google to pay Klarna's PriceRunner unit approximately SEK 14.3 billion (around $1.97 billion with interest) in antitrust damages arising from the same underlying conduct, finding that PriceRunner had suffered harm from Google's years-long favouring of its own comparison-shopping service – the largest competition damages award in Swedish history, and the first instance of private litigation translating the harm documented in settings like this paper's into a quantified award (Google may still appeal).[21] As in 2017, the Commission ordered Google to bring the conduct to an end within a fixed compliance window, once again leaving the specific redesign to Google's discretion.

This recurrence, under a legal framework explicitly designed to prevent it, is consistent with our central finding: naming and prohibiting self-preferencing is not sufficient when the regulated firm retains control over the granular design choices – ranking algorithms, interface layout, the visual prominence of boxes and units – through which compliance is ultimately implemented. It suggests that regulators have not yet fully internalized the lesson of the 2017 case, or that closing this gap requires enforcement design beyond simply strengthening ex-ante prohibitions. Concretely, our results point to three considerations for effective platform regulation: (1) ex-post monitoring mechanisms that provide visibility into platform design choices, rather than relying solely on self-reported compliance; (2) observable metrics for remedy success that are defined independently of platform implementation; (3) anticipatory specification of design constraints to limit discretion. Whether the 2026 remedy proves more durable than its 2017 predecessor is an open empirical question, and one that the difference-in-differences design developed in this paper is directly suited to answer once the redesigned SERP takes effect.

Our findings also offer a template for interpreting the DMA's ongoing self-preferencing enforcement. The 2026 finding of non-compliance again leaves Google discretion over the specific redesign, much as in 2017. Our results suggest that assessing this new remedy should look beyond whether third parties gain some form of formal access, since that is where the 2017 remedy appeared compliant while failing in practice. A more informative approach,

[20] European Commission, "Commission fines Google €890 million for breaches of the Digital Markets Act," 23 July 2026. https://digital-markets-act.ec.europa.eu/commission-fines-google-eur890-million-breaches-digital-markets-act-2026-07-23_en.

[21] Bloomberg, "Google Ordered to Pay Klarna Nearly $2 Billion in Antitrust Lawsuit," 1 July 2026. https://www.bloomberg.com/news/articles/2026-07-01/google-ordered-to-pay-klarna-nearly-2-billion-in-antitrust-suit.

following the three dimensions examined here, would track whether affected third parties' aggregate visibility and traffic actually recover, whether any new channel translates into a real improvement rather than shifting demotion elsewhere, and whether smaller rivals are disproportionately burdened relative to larger incumbents. Ultimately, the key question is not whether the adopted design formally opens the door to third parties, but whether it functions as a genuine gateway to competition, rather than a nominal channel that leaves the underlying competitive dynamics unchanged.

The Google Shopping case illustrates a challenge that current and future regulation may face: when dominant platforms retain control over remedy implementation, enforcement can become difficult to monitor and verify. This suggests that regulators might consider two approaches: either exercising more direct control over remedy implementation, or remedies must be designed to work despite platform discretion.

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A Appendix A

Table 7: Attrition and Panel Coverage

Variable	N	Mean	SD	Median
Complete coverage (39 CSS)				
Rank	2886	834,711	4,197,360	31,740

Continued on next page

Variable	N	Mean	SD	Median
Organic keywords	2886	213,806	673,021	33,542
Organic traffic	2886	643,805	2,635,962	12,354
AdWords traffic	2886	55,789	597,914	100
Near-complete coverage				
Rank	216	2,478,798	7,261,384	502,914
Organic keywords	216	5,624	16,058	216
Organic traffic	216	11,032	37,492	217
AdWords traffic	216	1,329	7,140	0
Partial coverage (15–30 periods)				
Rank	76	8,695,877	11,884,456	2,471,165
Organic keywords	76	1,712	6,489	55
Organic traffic	76	294	477	17
AdWords traffic	76	919,357	7,722,556	0
Marginal coverage (<15 periods)				
Rank	22	11,016,094	4,921,030	10,508,448
Organic keywords	22	2.1	2.4	1
Organic traffic	22	0.45	1.47	0
AdWords traffic	22	0	0	0
Total				
Rank	3200	1,202,387	5,013,077	51,277
Organic keywords	3200	193,247	642,185	26,643
Organic traffic	3200	581,383	2,510,418	8,062
AdWords traffic	3200	72,239	1,318,439	100

Notes: Websites are classified according to the number of periods in which they appear in the dataset. Complete coverage corresponds to websites observed in all periods. Near-complete, partial and marginal groups include websites observed in progressively fewer periods.

Table 8: Event Study Results (Winsorized Outcomes)

	(1) Organic Traffic	(2) Rank	(3) AdWords Traffic
EU × Eventtime ($t = -19$)	0.078	-0.033	-0.021
	(0.266)	(0.191)	(0.845)
EU × Eventtime ($t = -18$)	0.124	-0.098	0.158
	(0.261)	(0.179)	(0.859)
EU × Eventtime ($t = -17$)	0.104	-0.066	-0.085
	(0.247)	(0.185)	(0.841)
EU × Eventtime ($t = -16$)	0.098	-0.096	0.180
	(0.249)	(0.177)	(0.800)
EU × Eventtime ($t = -15$)	0.050	-0.052	-0.778
	(0.257)	(0.180)	(0.819)
EU × Eventtime ($t = -14$)	0.042	-0.052	-0.164
	(0.255)	(0.176)	(0.809)
EU × Eventtime ($t = -13$)	0.009	-0.025	-0.659

	(1) Organic Traffic	(2) Rank	(3) AdWords Traffic
	(0.247)	(0.170)	(0.798)
EU × Eventtime ($t = -12$)	0.047	-0.021	-1.155
	(0.242)	(0.175)	(0.807)
EU × Eventtime ($t = -11$)	-0.036	0.051	-1.475*
	(0.241)	(0.173)	(0.850)
EU × Eventtime ($t = -10$)	-0.089	0.095	-1.323*
	(0.218)	(0.170)	(0.742)
EU × Eventtime ($t = -9$)	-0.149	0.046	-1.016
	(0.199)	(0.134)	(0.789)
EU × Eventtime ($t = -8$)	-0.107	0.030	-0.545
	(0.192)	(0.135)	(0.777)
EU × Eventtime ($t = -7$)	-0.182	0.070	-0.627
	(0.192)	(0.132)	(0.771)
EU × Eventtime ($t = -6$)	-0.005	-0.054	-1.186
	(0.168)	(0.119)	(0.848)
EU × Eventtime ($t = -5$)	0.019	-0.112	-0.874
	(0.162)	(0.113)	(0.940)
EU × Eventtime ($t = -4$)	0.091	-0.118	-0.637
	(0.160)	(0.112)	(0.670)
EU × Eventtime ($t = -3$)	0.055	-0.062	-0.925
	(0.144)	(0.104)	(0.572)
EU × Eventtime ($t = -2$)	0.047	0.023	-0.288
	(0.077)	(0.053)	(0.193)
Reference period: event time $t = -1$ is omitted.			
EU × Eventtime ($t = 0$)	-0.048	0.037	0.459
	(0.055)	(0.043)	(0.535)
EU × Eventtime ($t = 1$)	0.054	0.000	1.112
	(0.095)	(0.065)	(0.970)
EU × Eventtime ($t = 2$)	0.023	-0.007	0.284
	(0.113)	(0.077)	(0.636)
EU × Eventtime ($t = 3$)	-0.090	0.129	0.094
	(0.118)	(0.095)	(0.591)
EU × Eventtime ($t = 4$)	-0.238	0.135	0.152
	(0.174)	(0.124)	(0.611)
EU × Eventtime ($t = 5$)	-0.550**	0.360**	0.465
	(0.240)	(0.171)	(0.601)
EU × Eventtime ($t = 6$)	-0.627*	0.445**	0.366
	(0.327)	(0.222)	(0.529)
EU × Eventtime ($t = 7$)	-0.607*	0.452*	-0.227
	(0.353)	(0.236)	(0.495)
EU × Eventtime ($t = 8$)	-0.778**	0.537**	-0.164
	(0.339)	(0.241)	(0.589)
EU × Eventtime ($t = 9$)	-0.988***	0.653***	-0.207
	(0.355)	(0.244)	(0.433)
EU × Eventtime ($t = 10$)	-1.044***	0.766***	-0.491
	(0.396)	(0.249)	(0.544)
EU × Eventtime ($t = 11$)	-1.277***	0.895***	-0.059
	(0.433)	(0.269)	(0.494)

	(1) Organic Traffic	(2) Rank	(3) AdWords Traffic
EU × Eventtime ($t = 12$)	-1.696***	1.148***	-0.006
	(0.391)	(0.261)	(0.501)
EU × Eventtime ($t = 13$)	-1.772***	1.213***	-0.320
	(0.381)	(0.262)	(0.396)
EU × Eventtime ($t = 14$)	-1.823***	1.232***	-0.230
	(0.375)	(0.254)	(0.529)
EU × Eventtime ($t = 15$)	-1.855***	1.283***	-0.660
	(0.419)	(0.285)	(0.419)
EU × Eventtime ($t = 16$)	-1.573***	1.056***	-0.554
	(0.460)	(0.318)	(0.428)
EU × Eventtime ($t = 17$)	-1.594***	1.095***	-1.240***
	(0.494)	(0.345)	(0.385)
EU × Eventtime ($t = 18$)	-1.689***	1.179***	-1.183***
	(0.489)	(0.350)	(0.333)
Knowledge Panel	0.000018*	-0.000012*	0.000008**
	(0.000009)	(0.000007)	(0.000004)
Constant	8.893***	10.343***	5.218***
	(0.122)	(0.089)	(0.298)
Observations	2,886	2,886	2,886
Website-Country FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Clusters (Website-Country)	76	76	76
R^2	0.905	0.920	0.793

Table 9: Effect of Carousel Features on Organic Traffic to CSS

	(1) Organic Traffic
log(Carousel)	−0.140***
	(0.032)
log(Rank)	−1.599***
	(0.057)
log(AdWords Traffic)	0.056*
	(0.029)
Website-Country FE	Yes
Time FE	Yes
Observations	1,680

Notes: Robust standard errors clustered at domain-country level. *** p<0.01, ** p<0.05, * p<0.10.

A.1 Robustness checks

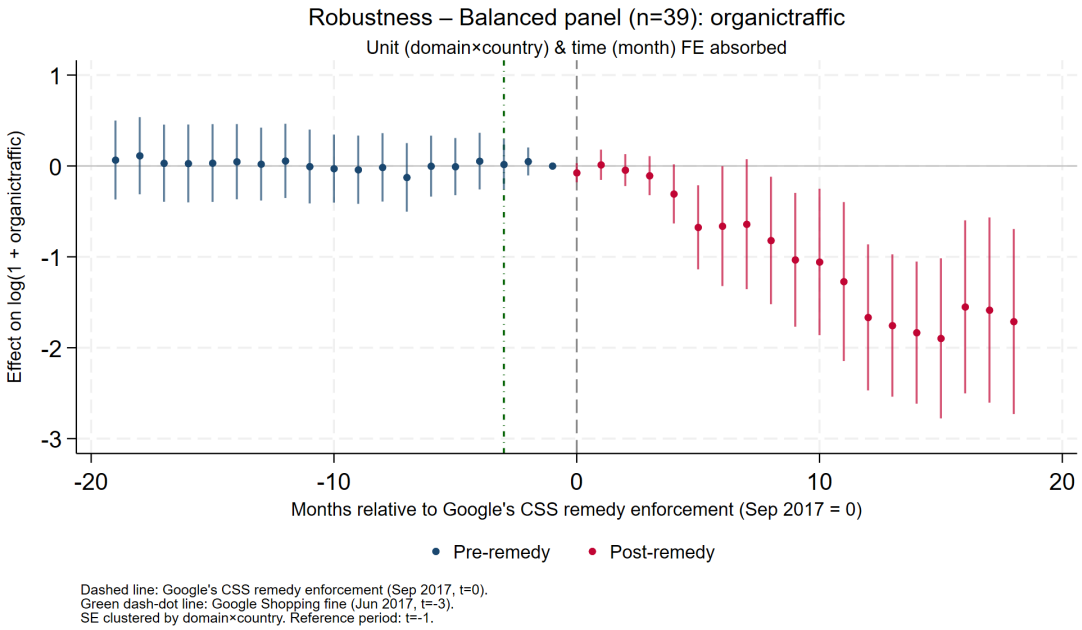


Figure 6: Event-study estimates of the effect of Google's redesign on organic search traffic, strictly balanced panel. The vertical dashed line indicates the implementation of Google's redesign.

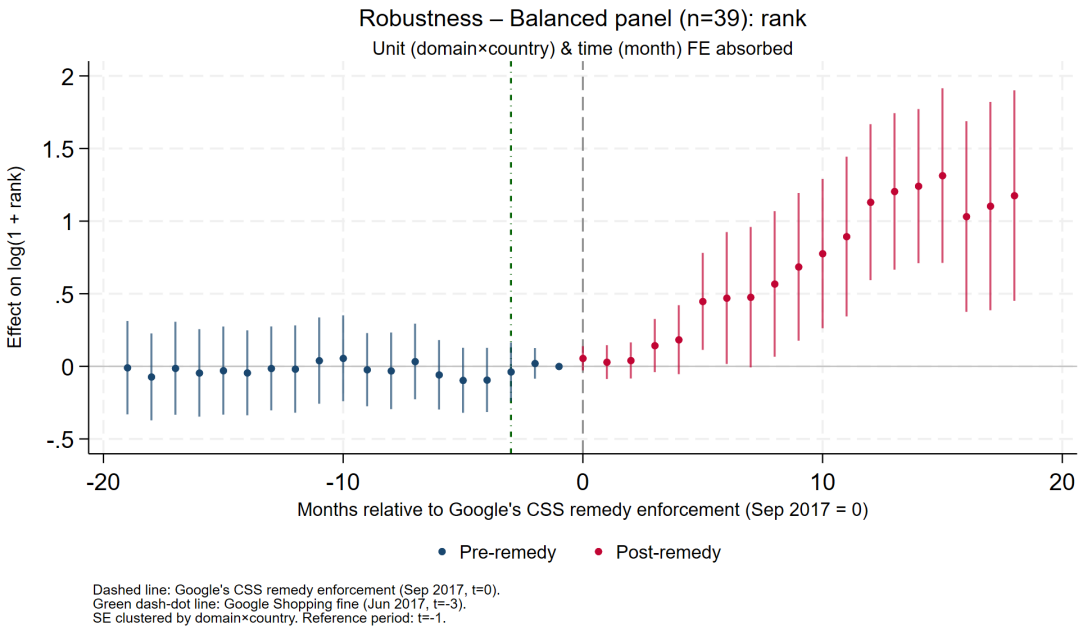


Figure 7: Event-study estimates of the effect of Google's redesign on ranking positions, strictly balanced panel. The vertical dashed line indicates the implementation of Google's redesign.

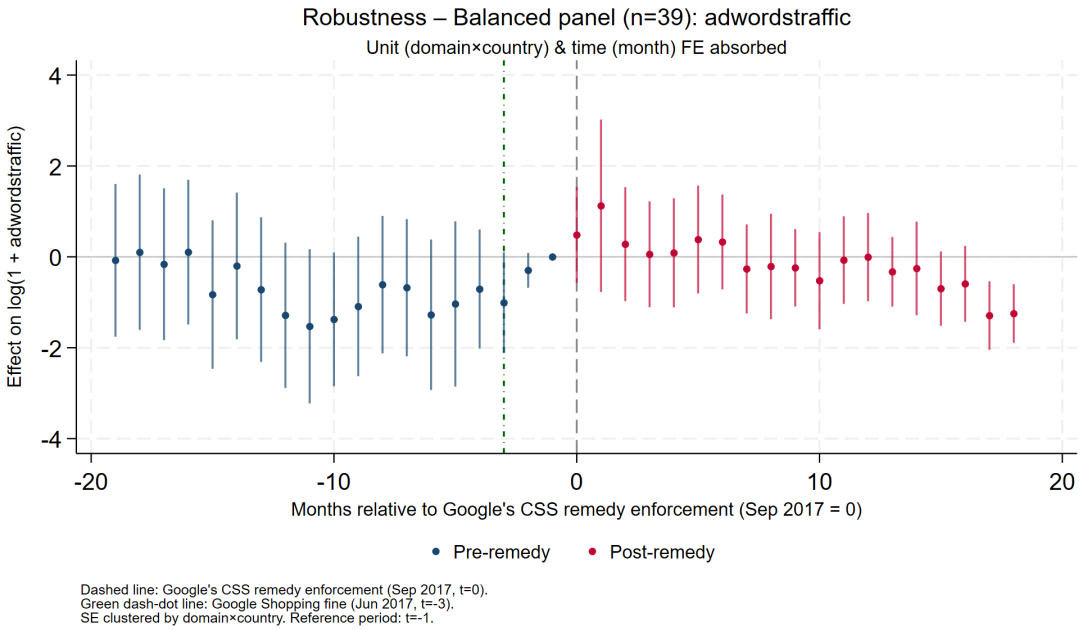


Figure 8: Event-study estimates of the effect of Google's redesign on paid search traffic, strictly balanced panel. The vertical dashed line indicates the implementation of Google's redesign.

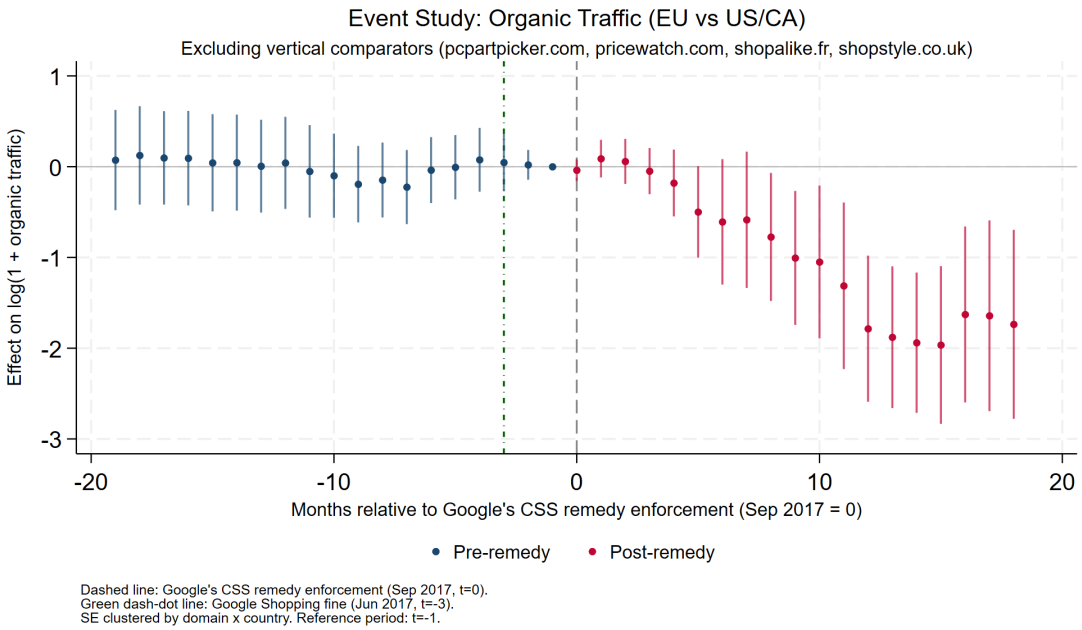


Figure 9: Event-study estimates of the effect of Google's redesign on organic traffic, excluding the four vertical CSS (PCPartPicker.com, PriceWatch.com, Shopalike.fr, ShopStyle.co.uk). The vertical dashed line indicates the implementation of Google's redesign.

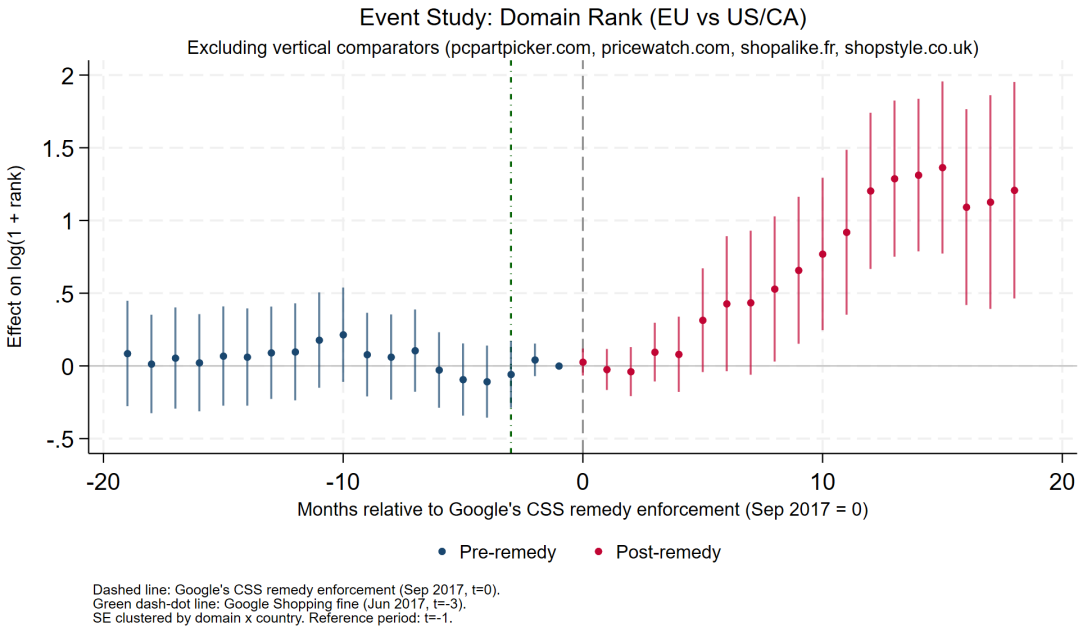


Figure 10: Event-study estimates of the effect of Google's redesign on ranking positions, excluding the four vertical CSS. The vertical dashed line indicates the implementation of Google's redesign.

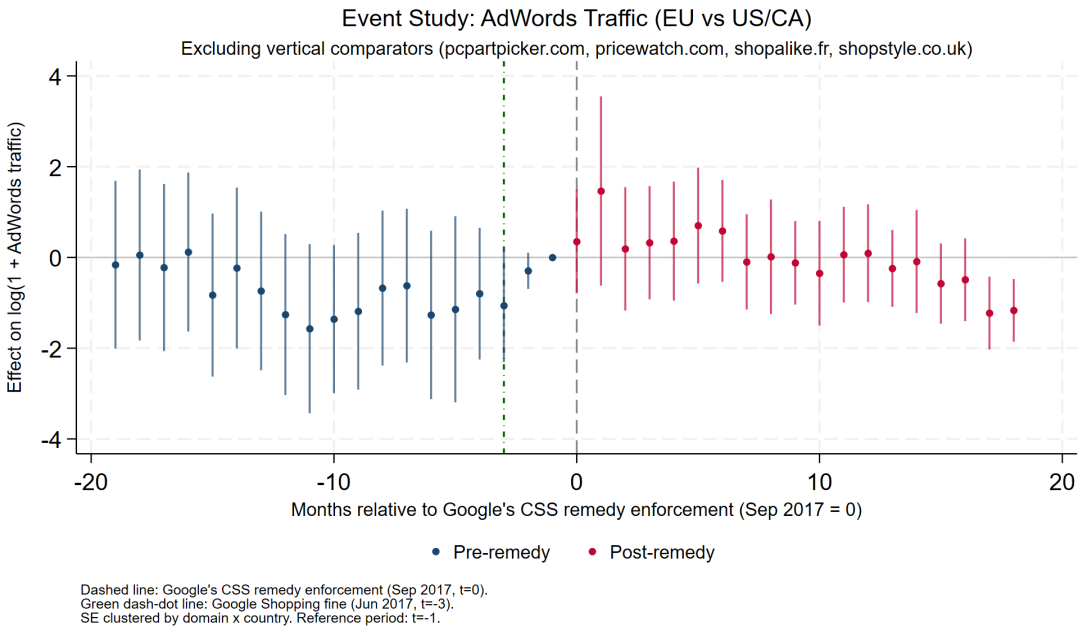
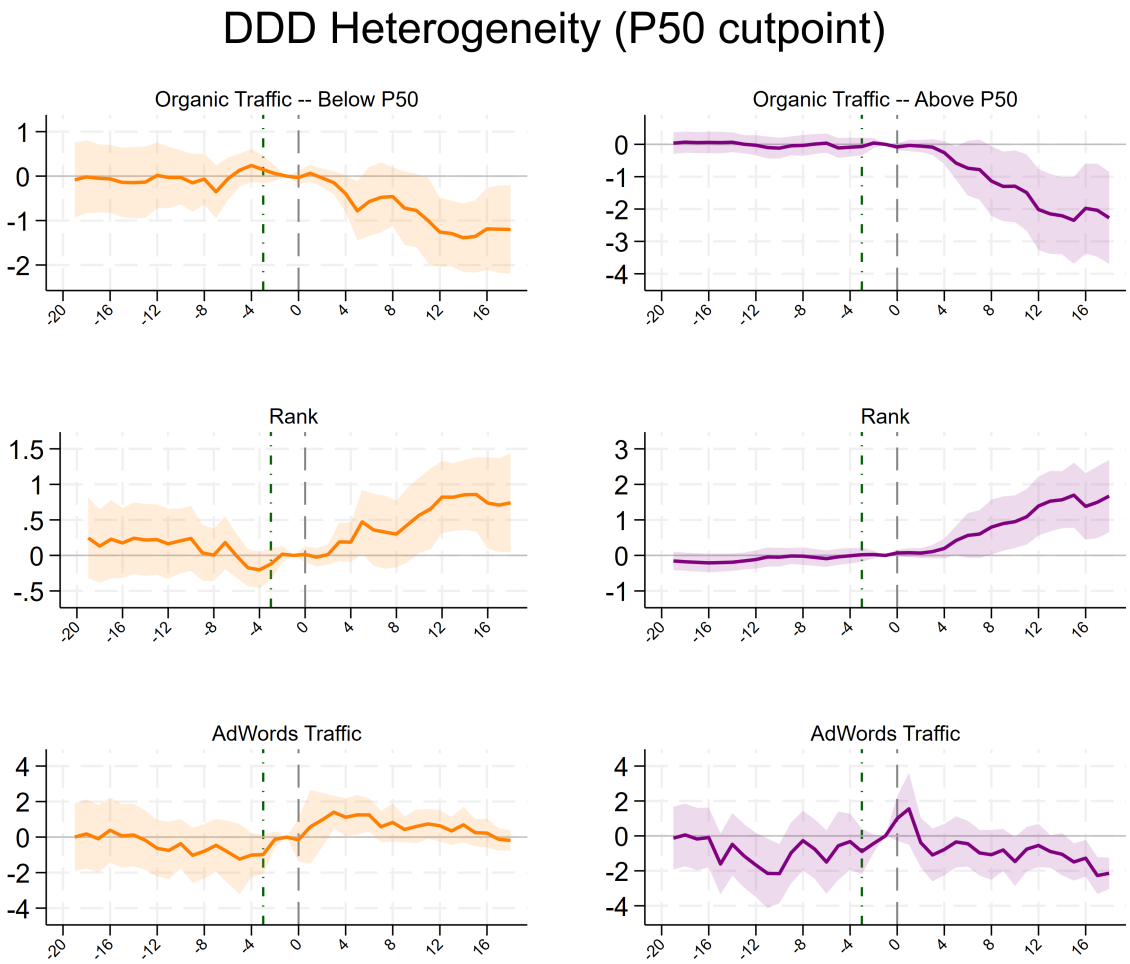


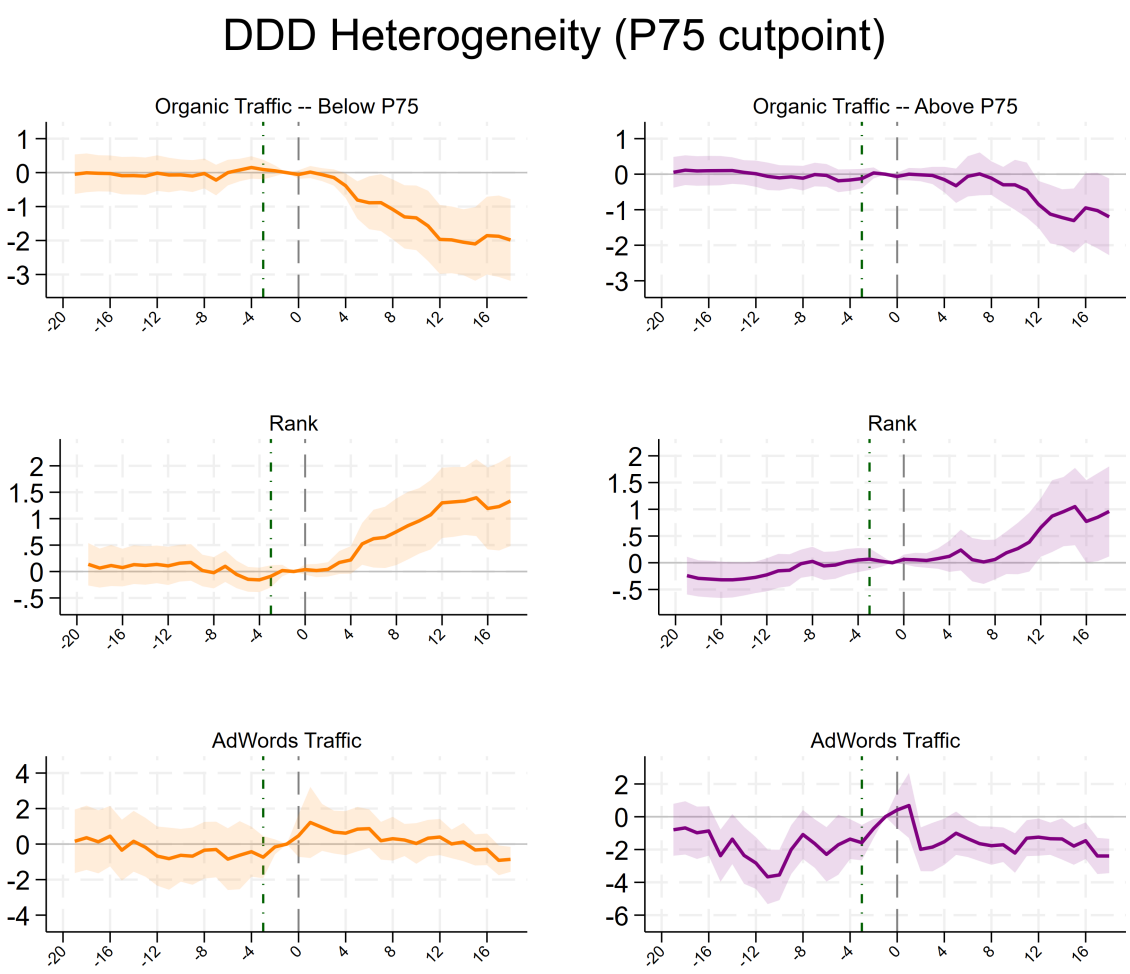
Figure 11: Event-study estimates of the effect of Google's redesign on paid search traffic, excluding the four vertical CSS. The vertical dashed line indicates the implementation of Google's redesign.

A.2 Heterogeneity Analysis



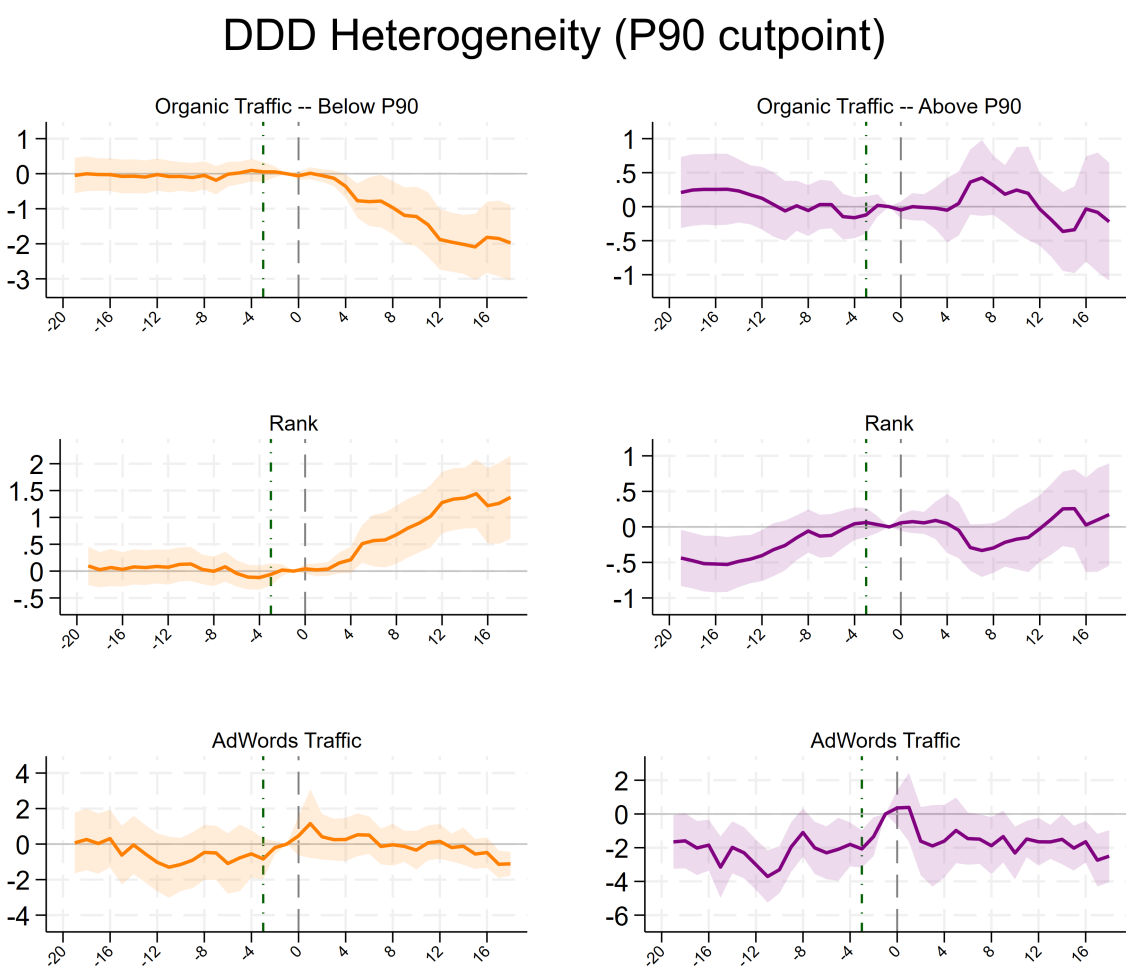
Full-sample DDD. Left column: below P50 (median). Right column: above P50.
Rows (top-bottom): Organic Traffic, Rank, AdWords Traffic. Unit and time FE absorbed. SE clustered by domain x c
Green dash-dot: Google Shopping fine (Jun 2017, t=-3). Dashed: CSS remedy (Sep 2017, t=0).

Figure 12: Triple-difference event-study estimates of Google's redesign using the median (P50) cutoff for pre-treatment traffic. Rows (top to bottom): organic traffic, ranking position, and paid (AdWords) traffic. Columns: CSS below the median (left) and above the median (right). The vertical dashed line indicates the implementation of Google's redesign; the dash-dot line marks the Commission's June 2017 fine.



Full-sample DDD. Left column: below P75 (cutoff). Right column: above P75.
Rows (top-bottom): Organic Traffic, Rank, AdWords Traffic. Unit and time FE absorbed. SE clustered by domain x c
Green dash-dot: Google Shopping fine (Jun 2017, t=-3). Dashed: CSS remedy (Sep 2017, t=0).

Figure 13: Triple-difference event-study estimates of Google's redesign using the 75th percentile (P75) cutoff for pre-treatment traffic. Rows (top to bottom): organic traffic, ranking position, and paid (AdWords) traffic. Columns: CSS below (left) and above (right) the threshold.



Full-sample DDD. Left column: below P90 (cutoff). Right column: above P90.
Rows (top-bottom): Organic Traffic, Rank, AdWords Traffic. Unit and time FE absorbed. SE clustered by domain x c
Green dash-dot: Google Shopping fine (Jun 2017, t=-3). Dashed: CSS remedy (Sep 2017, t=0).

Figure 14: Triple-difference event-study estimates of Google's redesign using the 90th percentile (P90) cutoff for pre-treatment traffic. Rows (top to bottom): organic traffic, ranking position, and paid (AdWords) traffic. Columns: CSS below (left) and above (right) the threshold.

A.3 Within-Country Validation: Small vs. Large CSS

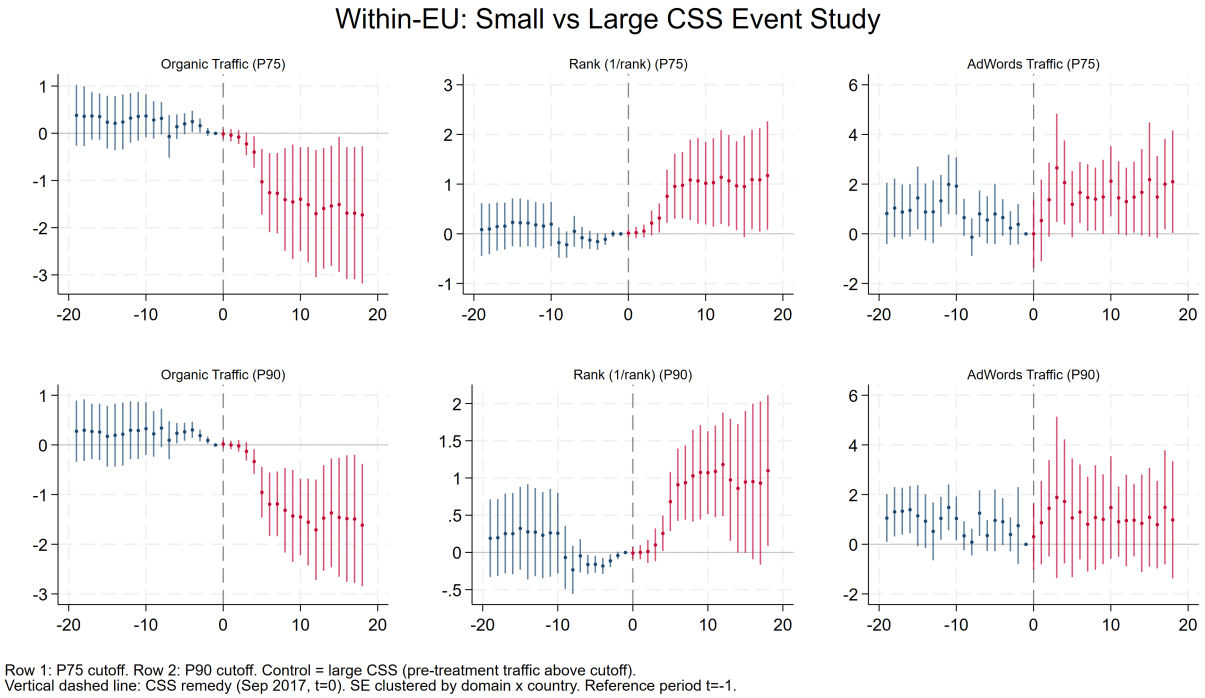


Figure 15: Within-EU event study: small CSS vs. large CSS, European countries only, for organic traffic, ranking position, and paid (AdWords) traffic. Top row uses the 75th percentile cutoff for pre-treatment traffic; bottom row uses the 90th percentile cutoff. Large CSS serve as an implicit control group in each panel. The vertical dashed line indicates the implementation of Google's redesign (September 2017).

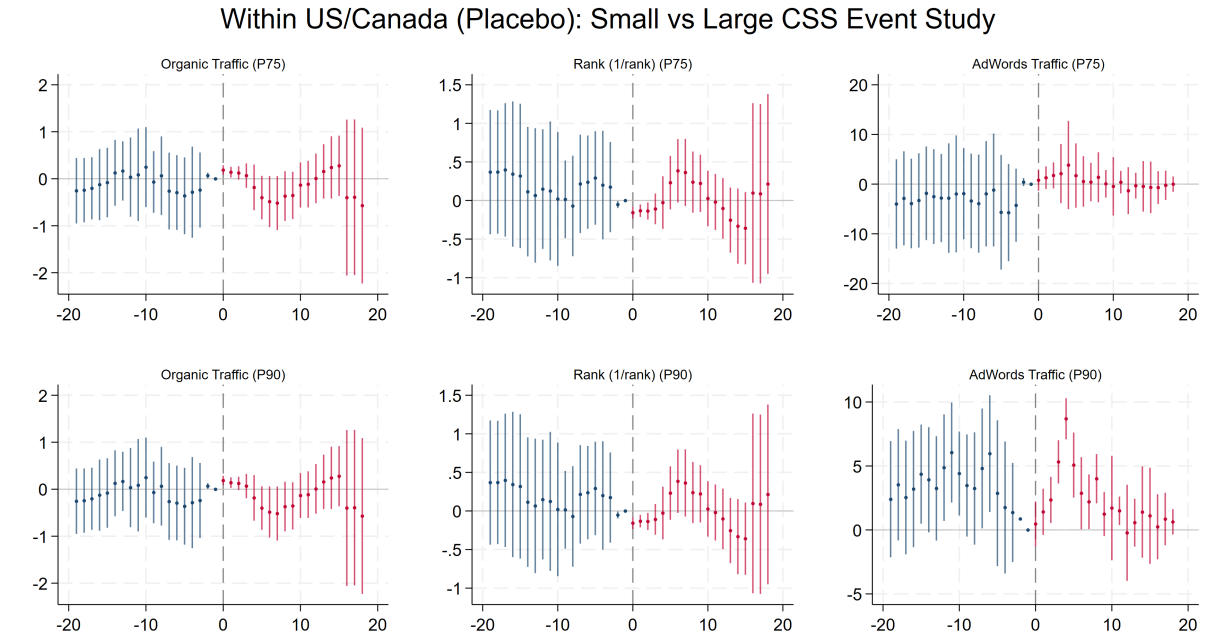


Figure 16: Placebo: identical small-vs-large CSS event study within the United States and Canada, where Google implemented no equivalent redesign, for the same three outcomes and two cutoffs as Figure 15. The post-treatment gap remains statistically indistinguishable from zero throughout for organic traffic and ranking position, in contrast to the treated-country results.

B Appendix B

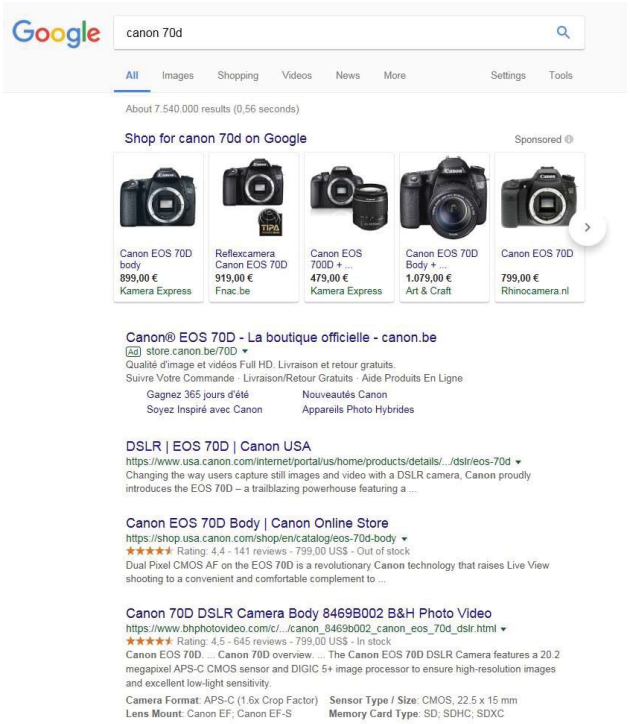


Figure 17: Illustration of a Google "One Box" (screenshot from www.google.be, 21 June 2017, reproduced from the European Commission's Antitrust Procedure). The boxed row above the organic results, "Shop for canon 70d on Google," is Google Shopping's One Box: a visually prominent placement for one of Google's own specialized services, embedded directly within the general search results page above the ordinary blue-link organic results shown further down.

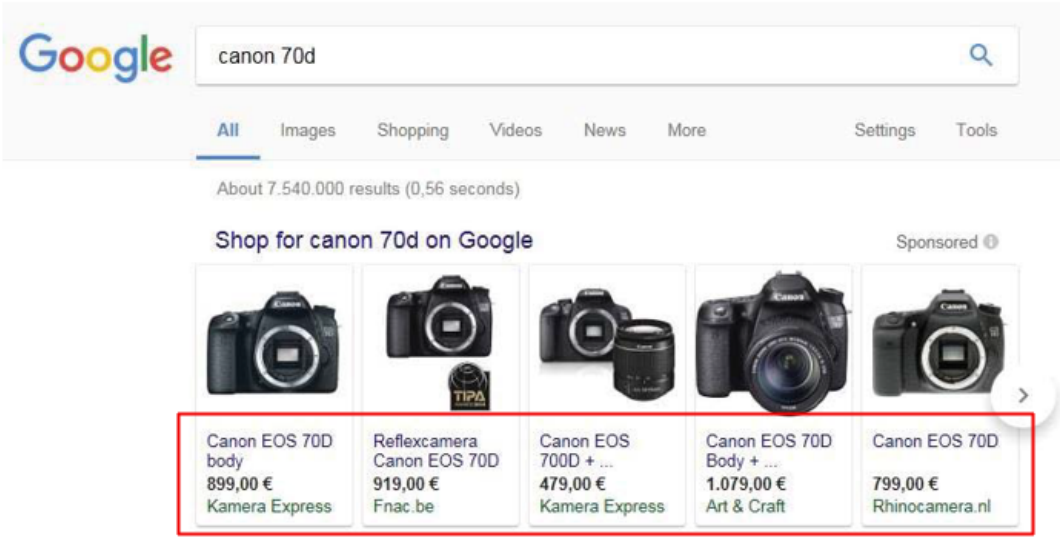


Figure 18: Google's Shopping Unit before the remedy (screenshot from www.google.be, 21 June 2017, reproduced from the European Commission's Antitrust Procedure). The header "Shop for canon 70d on Google" identifies this row of five tiles as Google's own Shopping placement, positioned above the organic search results. The product title and price on each tile – the tiles' only clickable element, highlighted here in red – link directly to a merchant's website (e.g. Kamera Express, Fnac.be, Art & Craft). No independent CSS (e.g. Kelkoo, idealo) is named anywhere in the unit: at this stage rival CSS had no means of appearing within this placement and could reach users only through the organic results further down the page.

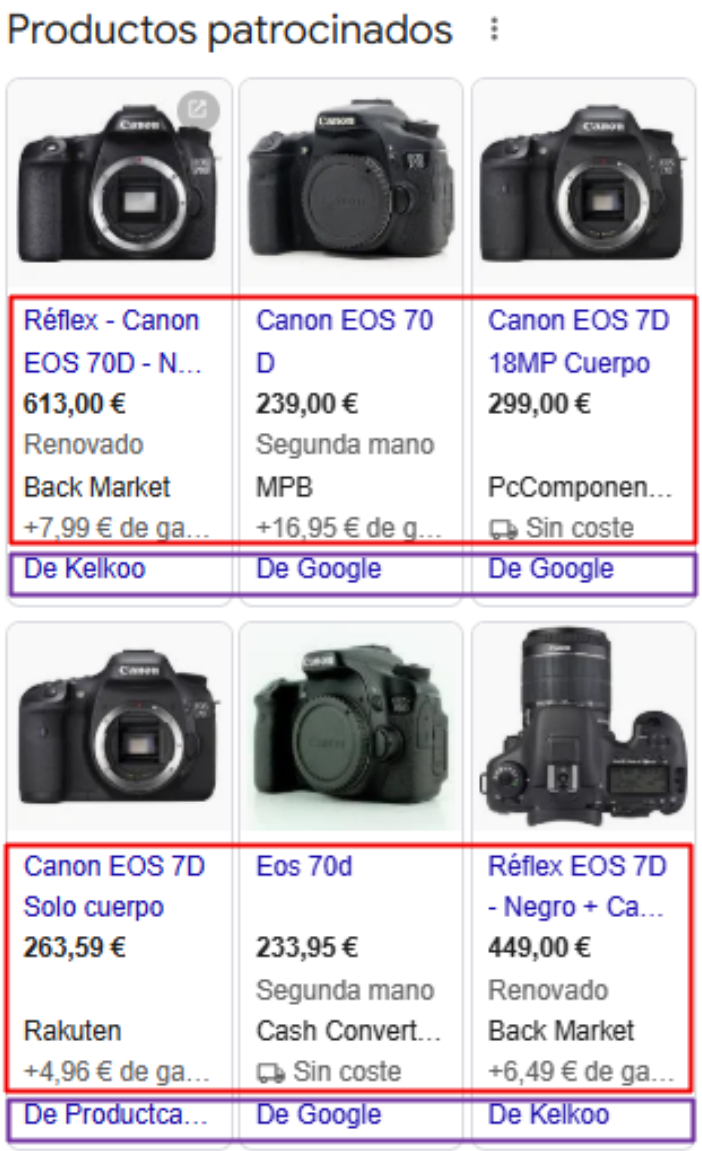


Figure 19: Google's Shopping Unit after the remedy (screenshot from www.google.es, 9 February 2026; author's own). As in Figure 18, the header ("Productos patrocinados," Sponsored products) marks this grid as Google's Shopping placement. Each tile now has two separately clickable areas, marked here in different colors. The area highlighted in red – the product image, title, and price, by far the largest part of the tile – is a link to the merchant's website, exactly as before the remedy. The small line beneath it, highlighted in purple ("De Kelkoo," "De Productcaza," "De Google"), is a separate link: clicking it takes the user to the CSS that supplied the listing (or, when the line reads "De Google," to no CSS at all). Four of the six tiles shown here are in fact sourced from a rival CSS, so a click anywhere on the tile except that one small purple-highlighted line still bypasses the CSS and lands the user on the merchant's page.

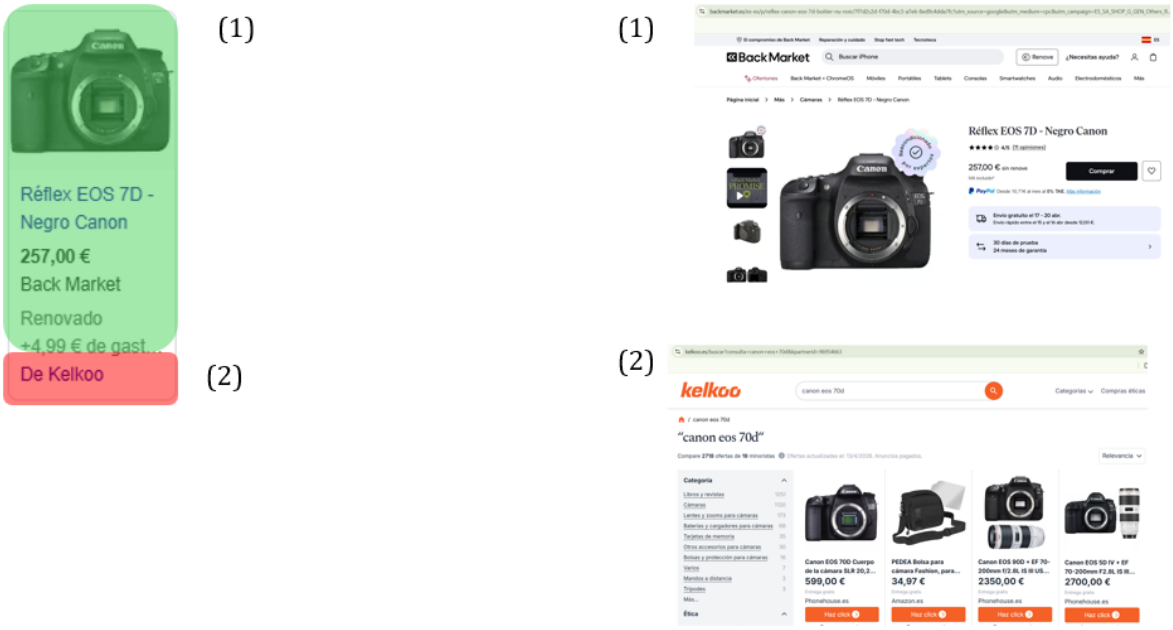


Figure 20: Where Shopping Unit clicks lead. Clicking the product tile (1) routes the user to the merchant's website (Back Market); clicking the small "by Kelkoo" attribution label (2) instead routes the user to the participating CSS's own comparison page (Kelkoo). The vast majority of user clicks fall on the much larger product tile rather than the small attribution label, illustrating mechanically why participating CSS receive so little of the resulting traffic. Screenshot taken 13 April 2026.

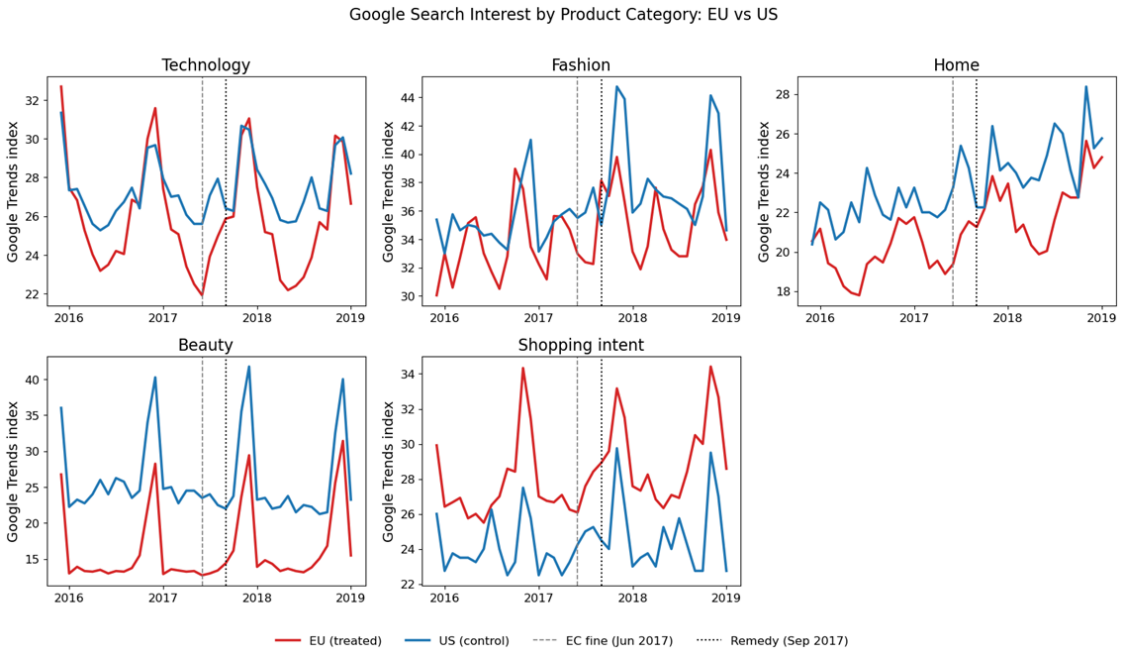


Figure 21: Google Search Interest by Product Category: EU vs. US (2016–2019). Search interest indices are constructed from Google Trends data covering more than 50 product categories representative of CSS activity. The dashed vertical line indicates the EC fine (June 2017) and the dotted vertical line the redesign implementation (September 2017). Red: EU (treated); Blue: US (control).